60. Which of the following statements is true of the respective mass and weight of an astronaut on Venus, where the acceleration due to gravity is 8.8 m/s^2 compared to that of the Earth's?
    A. Same mass and more weight
    B. Less mass and the same weight
    C. Less mass and more weight
    D. Same mass and less weight

BEST WISHES

AO/GA/RA/BSN/AT/FS

MYSTIC

MYSTIC VENTURES

UNIVERSITY OF CAPE COAST

COLLEGE OF AGRICULTURE AND NATURAL SCIENCES

SCHOOL OF PHYSICAL SCIENCES

DEPARTMENT OF PHYSICS

END OF FIRST SEMESTER EXAMINATION: 2020/2021

PH
GENERAL PHYSI[illegible] THEORY)

APRIL, 2021

TIME: 2 Hours

MYSTIC

ANSWER ALL QUESTIONS. ***For each question 1-60, read all the questions carefully and choose the most appropriate answer from alternatives lettered A-D. Each question carries 1 mark***

SOME CONSTANTS:

Quantity	Symbol	Value
Acceleration due to gravity	g	$9.8\ ms^{-2}$
Avogadro's number	N_A	$6.022\ 141\ 99 \times 10^{23}\ mol^{-1}$
Boltzmann's constant	k	$1.380\ 6503 \times 10^{-23}\ J/K$
Electron charge magnitude	e	$1.602\ 176\ 462 \times 10^{-19}\ C$
Permeability of free space	μ_o	$4\pi \times 10^{-7}\ T.m/A$
Permittivity of free space	ε_o	$8.854 \times 10^{-12}\ C2/(N.m^2)$
Plank's constant	h	$6.626 \times 10^{-34}\ J.s$
Mass of electron	m_e	$9.109 \times 10^{-31}\ J.s$
Mass of neutron	m_n	$1.674 \times 10^{-27}\ kg$
Mass of proton	m_p	$1.672 \times 10^{-27}\ kg$
Speed of light in vacuum	c	$2.997 \times 10^{8}\ m/s$
Universal gravitational constant	G	$6.673 \times 10^{-11}\ N.m^2/kg^2$
Universal gas constant	R	$8.314\ J/(mol.K)$
Coulomb's constant	k	$8.99 \times 10^{9}\ N.m^2/C^2$

1. Which of the following expressions is correct for the fundamental unit of work?
   A. 1 Joule = 1 Newton x 1 ohm
   B. 1 Joule = 1 Newton x 1 second
   C. 1 Joule = 1 Newton x 1 metre
   D. 1 Joule = 1 Newton x 1 kilogram
2. Which of the following quantities has a dimension MLT^{-1}? [All parameters have their usual meanings.]
   A. Momentum
   B. Acceleration
   C. Velocity
   D. Force

56. How large a force is necessary to stretch a 2.0 mm diameter copper wire by 1.0%? [Take *Young's Modulus*, $Y = 11 \times 10^{10}\ N/m^2$].
   A. 2.1 kN
   B. 3.5 kN
   C. 6.9 kN
   D. 11.0 kN

57. Which of the following is a scalar quantity?
   A. Displacement
   B. Momentum
   C. Pressure
   D. Impulse

58. The unit of thermal conductivity is_____.
   A. $J\ m\ s\ K^{-1}$
   B. $J^2\ m\ s\ K^{-1}$
   C. $J\ m^{-1}\ s^{-1}\ K^{-1}$
   D. $J^{-1}\ m^{-1}\ s^{-1}\ K^{-2}$.

59. Two vectors are considered to be equal if they have same magnitude and_____.
   A. same direction
   B. different direction
   C. positive direction
   D. negative direction

52. A boy stands on top of a cliff of height h, and throws a ball vertically upwards with a speed of u. The maximum height H reached by the ball above the ground is _____.

A. $H = h - \frac{u^2}{2g}$

B. $H = h + \frac{u^2}{2g}$

C. $H = h + \frac{u^2}{g}$

D. $H = h + \frac{u}{2g}$

53. A child whirls a ball at the end of a rope, in a uniform circular motion. Which of the following statements is NOT true?

A. The speed of the ball is constant.
B. The velocity of the ball is constant.
C. The radius is constant
D. The magnitude of the ball's acceleration is constant.

54. Oscillations in air become damped due to _____.

A. tangential forces
B. viscous forces
C. normal forces
D. parallel forces

55. The stress on a wire that supports a load depends on all of the following factors **except**_____.

A. the length of the wire
B. the diameter of the wire
C. acceleration of gravity
D. the mass of the load

3. A tennis player hits a ball at $10ms^{-1}$ at an angle of 30° to the ground. What are the respective horizontal and vertical components of the velocity of the ball?

A. 8.7 ms^{-1}; 6.0 ms^{-1}
B. 9.7 ms^{-1}; 5.0 ms^{-1}
C. 9.7 ms^{-1}; 6.0 ms^{-1}
D. 8.7 ms^{-1}; 5.0 ms^{-1}

4. Arrange the following quantities in descending order of magnitude.
(i) 0.8 kg, (ii) 15 cg, (iii) 0.7×10^5mg, (iv) 1.1×10^{-8}Gg, (v) $8.7 \times 10^8 \mu g$

A. (iv), (i), (ii), (iii), (v)
B. (i), (v), (iii), (ii), (iv)
C. (v), (i), (iii), (iv), (ii)
D. (ii), (i), (iii), (v), (iv)

5. A 0.5 kg ball strikes a 0.25 kg ball at rest. After collision, the 0.5 kg ball moves with a velocity of 10 m/s and the other ball moves with a velocity of 14 m/s. State the type of collision between the balls and calculate the initial velocity of the 0.5 kg ball?

A. Elastic and 17 m/s
B. Inelastic and 17 m/s
C. Elastic and 34 m/s
D. Inelastic and 34 m/s

6. The displacement of a boat is given by the equation y = (1.2 m) cos (t / 2s + π/6). Find the position of the boat after 1s.

A. 0.620 m
B. 0.623 m
C. 0.624 m
D. 0.628 m

7. Assuming that m and t have different dimensions, which of the following expressions would be physically meaningful?

A. mt
B. $m+t$
C. $m-t$
D. $\frac{m+t}{m}$

Use the Van der Waals equation of state $\left(p+\frac{a}{V^2}\right)(V-b)=RT$ to answer questions **8** and **9**. [All parameters have their usual meanings].

8. What is the dimension of a?
   A. ML^5T^{-2}
   B. $M^5L^{-2}T$
   C. MLT^{-2}
   D. $M^{-3}L^5T^{-2}$
9. What physical quantity does b represent?
   A. Intermolecular force of attraction
   B. Volume of molecules of the gas
   C. Pressure exerted by the gas
   D. Atmospheric pressure
10. A rectangle has a length of 1.323 m and a width of 4.16 m. Using significant figure rules, what is the area of the rectangle?
    A. 5.50368 m^2
    B. 5.5037 m^2
    C. 5.504 m^2
    D. 5.50 m^2
11. A 5.00 kg mass moves in a circle of 0.700 m radius. If the maximum centripetal force that can be applied to the mass is 120 N, what is its maximum speed in m/s?
    A. 4.1
    B. 14.4
    C. 15.2
    D. 16.8

47. A person holds a bucket by applying a force of 10 N. He moves a horizontal distance of 5 m and then climbs up a vertical distance of 10 m. What is the total work done by him?
    A. 50 J
    B. 150 J
    C. 100 J
    D. 200 J
48. In a liquid-in-glass thermometer, the temperature of a body is determined by ____.
    A. the thickness of the glass-tube
    B. the length of the column of liquid
    C. the radiation from the object
    D. the resistance of the object
49. A simple harmonic oscillator takes 12.0 s to undergo five complete oscillations. Find the angular frequency.

    A. 3.52 rad/s
    B. 9.66 rad/s
    C. 2.62 rad/s
    D. 22.6 rad/s
50. The kinetic theory of ideal gas is based on assumption that_____.
    A. the gas molecules collide inelastically
    B. the gas molecules are like billiard balls
    C. the gas molecules collapse into liquid
    D. the collisions of the molecules do not obey the law of conservation of energy
51. The conversion between the Celsius (c) scale and the Kelvin (K) scale is_____.
    A. $T_c = T_k + 373$
    B. $T_K = \frac{9}{5}T_c + 32^\circ$
    C. $T_K = T_C + 273$
    D. $T_K = T_C - 273$

43. A simple harmonic oscillator takes 12.0 s to undergo five complete oscillations. Find the angular frequency.

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B. 9.66 rad/s
C. 2.62 rad/s
D. 22.6 rad/s

44. A body undergoing projectile motion follows a _____ path.

A. hyperbolic
B. parabolic
C. circular
D. linear

45. The dot product of a force vector and a displacement vector defines _____.

A. power by a constant force
B. energy by a varying force
C. work by a constant force
D. acceleration by a constant force

46. A centripetal force F is experienced by an object moving at a constant speed v in a horizontal circle of radius r. If the radius is quadrupled and the speed is doubled, what happens to the centripetal force?

A. It is increased by a factor of 2
B. It is decreased by a factor of 2
C. It remains unchanged
D. It is increased by a factor of $\sqrt{2}$

12. A tennis ball of mass 75 g moving at 4 m/s collides head-on with a golf ball of 80 g moving at 3 m/s in opposite to the direction of the tennis ball. If after collision, the tennis ball moved with 2.5 m/s, find the velocity of the golf ball after collision.

A. 2.0 m/s
B. 2.4 m/s
C. 4.4 m/s
D. 5.0 m/s

13. A ball of mass 0.30 kg moves on a frictionless horizontal surface. If two forces: $\mathbf{F_1} = 5\ \text{Cos}\ (-20^0)\mathbf{i} + 5\ \text{Sin}\ (-20^0)\mathbf{j}$ and $\mathbf{F_2} = 8\ \text{Cos}\ (60^0)\mathbf{i} + 8\ \text{Sin}\ (60^0)\mathbf{j}$ act on it, determine its acceleration in the x-direction.

A. 29.0 m/s
B. 17.0 m/s
C. 8.7 m/s
D. 5.2 m/s

14. If $\vec{A} = 12\hat{i} - 16\hat{j}$ and $\vec{B} = -24\hat{i} + 10\hat{j}$, what is the magnitude of the vector $\vec{C} = 2\vec{A} - \vec{B}$?

A. 22
B. 42
C. 64
D. 90

15. An automobile travelling along a straight road increases its speed from 30.0 m/s to 50.0 m/s in a distance of 180 m. The acceleration of the automobile is _____.

A. 9.00 ms^{-2}
B. 2.25 ms^{-2}
C. 0.44 ms^{-2}
D. 0.22 ms^{-2}

16. Which of the following is the biggest in magnitude?
    A. 1 Angstrom
    B. 1 Decameter
    C. 1 Gigameter
    D. 1 Megameter

17. A car is moving in a circle of radius 25 m at a constant speed of 12 m/s. What is the magnitude of the centripetal acceleration acting on the car?
    A. 0.48 m/s^2
    B. 2.08 m/s^2
    C. 5.76 m/s^2
    D. 6.40 m/s^2

18. The equation for the displacement of a train starting at $x = 0$ m is given by $x = \frac{1}{2}at^2 + bt^3$. What is the dimension of b?
    A. T^3
    B. LT^3
    C. LT^2
    D. LT^1

19. In an inelastic collision, _____ is/are conserved.
    A. momentum but not kinetic energy
    B. both momentum and kinetic energy
    C. kinetic energy but not momentum
    D. neither momentum nor kinetic energy

39. A 20 g particle moves in simple harmonic motion with a frequency of 15.0 Hz and an amplitude of 77.0 mm. Determine the maximum acceleration of the particle.
    A. 644.9 ms^{-2}
    B. 621.9 ms^{-2}
    C. 611.9 ms^{-2}
    D. 683.9 ms^{-2}

40. Consider an object sliding over a smooth horizontal surface. The work done by the force of gravity will be zero; because it_____.
    A. acts parallel to the displacement.
    B. has maximum component to the displacement.
    C. acts perpendicularly to the displacement.
    D. does not exist on the surface

41. A boy drags a load on a horizontal smooth floor with a force F = 4 N applied at an inclination of 60° to horizontal. If the load moves through a distance of 3 m. What is the work done by the force?
    A. 4 J
    B. 6 J
    C. 8 J
    D. 10

42. The work done in lifting a body vertically through a height h is _____.
    A. $mgh^{1/2}$
    B. ½ mv^2h
    C. mgh
    D. hgρ

34. The amount of heat energy required to raise the temperature of a unit mass of a substance by 1K defines the_____.
A. latent heat
B. heat capacity
C. specific latent heat
D. specific heat capacity

35. Calculate the energy required to raise the temperature of 2-kg of water from 20°C to 100°C. [Take specific heat capacity of water as 4200 $Jkg^{-1}\ °C^{-1}$].
A. 672000 J
B. 682000 J
C. 772000 J
D. 782000 J

36. Which of the following is NOT a mode of heat transfer?
A. Conduction
B. Radiation
C. Convection
D. Cooling

37. If two bodies collide, stick together and move with a common velocity, the_____.
A. kinetic energy will be conserved
B. momentum will be conserved
C. momentum will be reduced
D. potential energy will be greater than the kinetic energy

38. How much time will be required to perform 520 J of work at a rate of 20 W?
A. 26 s
B. 24 s
C. 20 s
D. 16 s

20. A particle is projected with velocity, v_o at an angle α to the horizontal. Which of the following expressions represents the range of the particle?
A. $\frac{v_o}{g}\sin(2\alpha)$
B. $\frac{V_o^2}{g}\sin(\alpha)$
C. $\frac{V_o^2}{g}\sin(2\alpha)$
D. $\frac{V_o^2}{2g}\sin(\alpha)$

21. The magnitude of a displacement vector **R** =175 m, and R makes an angle of 50.0° with the vertical. Find the x- and y-components of the vector.
A. (135 m, 134 m)
B. (112 m, 134 m)
C. (134 m, 215 m)
D. (134 m, 112 m)

22. If the velocity of a particle is doubled, its kinetic energy is_____.
A. doubled
B. halved
C. constant
D. quadrupled

23. A proton undergoing linear motion with a uniform acceleration of $6.0 \times 10^{12}\ m/s^2$ has an initial velocity 4.0×10^6 m/s. What is the velocity attained after covering a distance of 80 cm?
A. 6.3×10^6 m/s
B. 5.1×10^6 m/s
C. 4.8×10^6 m/s
D. 3.9×10^6 m/s

24. Which of the following is NOT a fundamental quantity?
    A. Amount of substance
    B. Electric current
    C. Resistance
    D. Mass

25. Which of the objects below has the greatest momentum? A_____
    A. 2-kg object moving at 0.2 m/s
    B. 1.5-kg object moving at 1 m/s
    C. 1-kg object moving at 4 m/s
    D. 0.5-kg object moving at 5 m/s.

26. What is the scalar product of the vectors $A = 2i + j - k$ and $B = i - 3j + 2k$?
    A. -1
    B. -3
    C. 3
    D. 7

27. Find the magnitude and direction of the vector $Q = 3i + 4j$.
    A. (5, 36.87°)
    B. (7, 36.87°)
    C. (5, 53.12°)
    D. (7, 53.12°)

28. If $P = (2, 4, 2)$, $Q = (-3, 1, -1)$ and $R = (1, 2, 3)$, Find $\mathbf{P} + 2\mathbf{Q} - \mathbf{R}$.
    A. (-5, 4, -3)
    B. (-3, -4, 5)
    C. (9, 4, -3)
    D. (-5, 8, -3)

29. Find the unit vector of $\mathbf{M} = 4i - 3j$.
    A. $\frac{4}{5}i + \frac{3}{5}j$
    B. $\frac{4}{25}i - \frac{3}{25}j$
    C. $\frac{4}{5}i - \frac{3}{5}j$
    D. $\frac{4}{25}i + \frac{3}{25}j$

30. Find the angle between the vectors $\mathbf{P} = (2, 1)$ and $\mathbf{B} = (4, 2)$.
    A. 30°
    B. 15°
    C. 5°
    D. 0°

31. What is the net force acting on an object moving with constant velocity?
    A. 0 N
    B. 1 N
    C. 10 N
    D. 100 N

32. The change in momentum of a body also defines its
    A. force
    B. power
    C. impulse
    D. Potential energy

33. A toy car of 2 kg was forced to move from rest to attain a speed of 5 m/s. Find the impulse on the toy car.
    A. 2.5 $kgms^{-1}$
    B. 7.0 $kgms^{-1}$
    C. 10.0 $kgms^{-1}$
    D. 15.0 $kgms^{-1}$

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DEPARTMENT OF PHYSICS

END OF FIRST SEMESTER EXAMINATION

PHY 101: GENERAL PHYSICS I

DURATION: 2 HOURS

DECEMBER, 2017

MYSTIC VENTURES

MYSTIC VENTURES

ANSWER ALL QUESTIONS

1. The following are all scalar quantities **except**
   A. Displacement
   B. Distance
   C. Speed
   D. Mass

2. A steel ball of mass 140 g rolls down a frictionless plane, inclined at 30° to the horizontal, with a uniform acceleration equal to
   A. 20.0 ms^{-2}
   B. 10.0 ms^{-2}
   C. 8.7 ms^{-2}
   D. 5.0 ms^{-2}

3. Which statement is **TRUE** about the unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$?
   A. Their directions are defined by a left-handed coordinate system.
   B. The angle between any two is 90 degrees.
   C. Each has a length of 1 m.
   D. If $\hat{i}$ is directed east and $\hat{j}$ is directed south, $\hat{k}$ points up out of the surface.

4. Find the resultant of three concurrent forces of 8 N, 12 N and 16 N acting at respective angles 30°, 60° and 90° relative to the +x-direction.
   A. 43.0 N, 067°
   B. 36.0 N, 180°
   C. 30.4 N, 090°
   D. 43.0 N, 247°

5. Given two vectors $\vec{M} = (3i - j + 2k)$ and $\vec{N} = (i + 2j - 5k)$. Find the angle between $\vec{M}$ *and* $\vec{N}$.
   A. 26°
   B. 38°
   C. 61°

   A. load hasn't exceeded elastic limit yet; so, Hooke's law applies
   B. load has already exceeded elastic limit and material has become plastic
   C. even plastic stage has passed and wire has snapped already
   D. the material does not extend further with applied load

59. Materials which elongate considerably and undergo plastic deformation before rupture are said to be
   A. brittle
   B. breakable
   C. ductile
   D. elastic

60. A body is said to be elastic if it
A. recovers its length after removing the applied stress
B. does not return to its original length after removing the applied stress
C. has a Young's modulus greater than 1 Nmm^{-2}
D. can support a large load without breaking

D. $3.4 \times 10^2 m/s$

54. Find the mass of the ethyl alcohol molecule C_2H_6O given that the atomic masses of H, C and O are 1.008, 12.01 and 16.00, respectively. [$1\ amu = 1.66 \times 10^{-27} kg$.]

A. $6.65 \times 10^{-26} kg$
B. $7.65 \times 10^{-26} kg$
C. $8.65 \times 10^{-26} kg$
D. $9.65 \times 10^{-26} kg$

55. Which statement below is ***NOT*** an assumption made in the molecular model of an ideal gas?

A. The average separation between molecules is large compared with the dimensions of the molecules.
B. The molecules undergo inelastic collisions with one another.
C. The forces between molecules are short range.
D. The molecules obey Newton's laws of motion.

56. Tensile strain is defined as

A. Force per unit area of cross-section
B. Extension per unit area of cross-section
C. Extension per unit length
D. Force per unit length

57. Hooke's law states that

A. extension is proportional to load when elastic limit has not been exceeded
B. extension is inversely proportional to load when elastic limit has not been exceeded
C. extension is independent of load when elastic limit is not exceeded
D. load is dependent on extension in the elastic region

58. At 'yield point' of a copper wire

D. 64°

6. A boy pulls a toy with a force F, inclined at an angle 35° to the horizontal. If the effective force pulling the toy along the horizontal floor is 28 N, what is the value of F?

A. 34.3 N
B. 29.2 N
C. 22.9 N
D. 16.1 N

7. Which of the following correctly represents a force of 13 N inclined at 67.5° to the +x-direction?

A. 12i + 5j
B. 12i – 5j
C. 5i – 12j
D. 5i + 12j

8. The prefix *femto* has a factor value of

A. 10^{-12}
B. 10^{12}
C. 10^{15}
D. 10^{-15}

9. The dimensions of power is

A. $ML^{-1}T$
B. $ML^{-1}T^{-2}$
C. $ML^{-2}T^{-1}$
D. ML^2T^{-3}

10. In order for two objects to have the same temperature, they must

A. be in thermal equilibrium.
B. be in thermal contact with each other.
C. have the same relative "hotness" or "coldness" when touched.
D. have all of the properties listed above.

11. The prefix for the factor 10^{12}

A. Pico
B. Nano
C. Tera
D. Mega

12. One number has three significant figures and another number has four significant figures. If these numbers are added, subtracted, multiplied, or divided, which operation can produce the greatest number of significant figures?
A. the addition
B. the subtraction
C. the multiplication
D. the division

13. John and Linda are arguing about the definition of density. John says the density of an object is proportional to its mass. Linda says the object's mass is proportional to its density and to its volume. Which one, if either, is correct?
A. They are both wrong.
B. John is correct, but Linda is wrong.
C. John is wrong, but Linda is correct.
D. They are both correct.

14. What is the volume (in m^3) of a sheet of length is 4.3 m, breadth is 5.4 m and thickness is 3.5 cm?
A. 0.813
B. 1.890
C. 23.229
D. 81.270

15. Systematic error which occurs due to poor calibration of an instrument can be corrected by
A. taking several readings
B. replacing instruments
C. taking mean of values
D. tabulating the values

C. Molecular motion stops
D. Liquid hydrogen freezes

50. Real gases show remarkable deviation from that of ideal gas behavior at
A. High temperature and low pressure
B. Low temperature and high pressure
C. High temperature and high pressure
D. Low temperature and low pressure

51. Estimate the mean free path of an air molecule of diameter 4.0 x 10^{-10} m at 273 K and one atmosphere (1.01 x 10^5 Pa). Given that $N_A = 6.02 \times 10^{23}$ and $R = 8.31 J/K$)

A. $5.2 \times 10^{-8}\, m$

B. $6.2 \times 10^{-9}\, m$

C. $7.2 \times 10^{-10}\, m$

D. $8.2 \times 10^{-11}\, m$

52. What is the translational kinetic energy of one mole of a gas?
A. $2/3(RT)$.
B. $1/2(RT)$
C. $2/1(RT)$
D. $3/2(RT)$

53. What is the average velocity of the molecules in a sample of oxygen at 100 ^{0}C? The mass of an Oxygen molecule is $5.3 \times 10^{-26} kg$ [Boltzmann's constant $k - 1.38 \times 10^{-23} J/K$.]

A. $6.4 \times 10^2 m/s$

B. $5.4 \times 10^2 m/s$

C. $4.4 \times 10^2 m/s$

3 atmospheres from a volume of 2 to 5 liters? [1 atmosphere = $1.013 \times 10^5 Pa$]

A. 912 J
B. 812 J
C. 712 J
D. 612 J

46. What is the Celsius equivalent of 80°F?
A. 36.7 °C
B. 26.7 °C
C. 16.7 °C
D. 46.7 °C

47. Determine the amount of heat that will be added to 3 kg of water to raise its temperature from 20 to 80 ° C. [specific heat capacity of water is 4.185 KJ/(kg. ° C)]

A. 453 KJ
B. 553KJ
C. 653 KJ
D. 753 KJ

48. The freezing point of water is 0 ° C. Under similar conditions, its melting point will be

A. slightly lower than 0 °C
B. equal to 0 ° C
C. slightly higher than 0 ° C
D. equal to 32 0 ° C

49. According to the kinetic theory of gases, at absolute zero of temperature
A. Water freezes
B. Liquid helium freezes

16. A golf ball leaves the golf club at an angle of 60° above the horizontal ground with a velocity of 40m/s. What is the maximum height that the ball can reach? [Take g = 10m/s²]
A. 40 m
B. 80 m
C. 60 m
D. 160 m

17. A catapult was used to project a stone at an angle of 45° to the horizontal with velocity of 20 m/s. Calculate the horizontal range of the stone. [Take g = 10m/s²]
A. 20 m
B. 40 m
C. 200 m
D. 400 m

18. What is the angular speed of a car if it moves with a uniform speed of 30m/s in a circle of radius 6 m?
A. 5 rad/ s
B. 10 rad/ s
C. 15 rad/ s
D. 30 rad/ s

19. A particle moves round a circular track of radius 0.4 m at 2 revolutions per second. Calculate the period T of the motion.
A. 0.5 s
B. 2.0 s
C. 2.5 s
D. 4.0 s

20. A mass suspended vertically from a spring makes 18 vibrations in 40 s. Find the frequency of the vibration.
A. 0.35 Hz
B. 0.45 Hz

MYSTIC

C. 1 24 Hz
D. 2.22 Hz

21. A vertical light spring stretches 25 cm when a mass of 0.50 kg hangs at its free end. What is the spring constant? [Take $g = 10 m/s^2$]

A. 5 N/ m
B. 10 N/ m
C. 15 N/ m
D. 20 N/ m

22. Find the instantaneous acceleration of an athlete at time t = 5 s, if he runs with velocity
$v = 3t^2\,i + 2t\,j$. [Leave your answer in vector notation]

A. $3\,i + 2\,j$
B. $6\,i + 2\,j$
C. $30\,i + 2\,j$
D. $32\,i$

23. A stonecutter's chisel has an edge area of 0.7 cm^2. If the chisel is struck with a force of 42 N, what is the pressure exerted on the stone?

A. 600 N/m^2
B. 30 000 N/m^2
C. 300 000 N/m^2
D. 600 000 N/m^2

24. A hydraulic lift raises a 2 000-kg automobile when a 500-N force is applied to the smaller piston. If the smaller piston has an area of 10 cm^2, what is the cross-sectional area of the larger piston?

A. 40 cm^2
B. 80 cm^2
C. 196 cm^2
D. 392 cm^2

41. An arrow is drawn back so that 50 Joules of potential energy is stored in the stretched bow and string. When released, the arrow will have a kinetic energy ____ Joules.

A. more than 50
B. equal to 50
C. less than 50
D. between 50 and 100

42. A man uses a horizontal force of 200 N to push a crate up an inclined plane 8 m long that is 20° above the horizontal. If Martin takes 12 seconds to push the crate up the plane, what is his output power?

A. 125 W
B. 166 W
C. 1504 W
D. 1600 W

43. Which of the following statements represents the first law of thermodynamics?

A. Net heat input = change in internal energy + net heat output
B. Net heat output = net work output + change in internal energy
C. Net heat output = change in internal energy + network output
D. Net heat input = change in internal energy + network output

44. The rate of heat flow through a slab of material depends on the following **except**?

A. The thickness of the slab
B. The cross-section area of the slab
C. The specific heat capacity
D. The temperature difference between the faces of the slab

45. How much work is done when a gas expands at constant pressure of

C. 2.0 ms^{-1}
D. 2.5 ms^{-1}

37. A job is done slowly, and an identical job is done quickly. Both jobs require the same amount of ____, but different amounts of ____.

A. energy, work
B. power, work
C. work, energy
D. work, power

38. A 50-kg platform diver hits the water below with a kinetic energy of 5000 Joules. The height (relative to the water) from which the diver jumped was approximately ____ meters.

A. 5
B. 10
C. 50
D. 100

39. A book is placed on a chair. Then a videocassette is placed on the book. The floor exerts a normal force

A. on all three.
B. only on the book.
C. only on the chair.
D. upwards on the chair and downwards on the book.

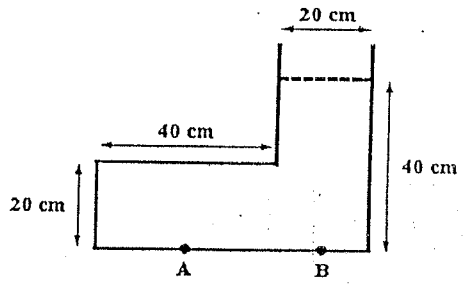
40. A hammer with 1.5 kg head is used to drive a nail into a wooden board. If the hammer head is moving at 3 m/s when it strikes the nail and the nail moves 10 mm into the board, find the average force the hammer head exert on the nail.

A. 450 kN
B. 13.5 kN
C. 4.5 kN
D. 1.35 kN

25. A wood block is placed on top of the ice in a large bowl half full of ice. The bowl is then filled to the brim with water, with the wood block riding on top of the ice. As the ice melts,

A. the density of the water decreases.
B. the water level falls below the rim.
C. the water level rises and water spills out of the bowl.
D. the water level does not change.

26. The figure below shows a container filled with water to the height shown. By comparing the pressure at A to the pressure at B, we find that :



A. $p_A = \frac{1}{4} p_B$.
B. $p_A = \frac{1}{2} p_B$.
C. $p_A = p_B$.
D. $p_A = 2p_B$.

27. Some species of whales can dive to depths of one kilometer. What is the total pressure they experience at this depth? ($\rho_{seawater}$ = 1 020 kg/m^3 and 1.01 x10^5 N/m^2 = 1 ATM, g = 9.8 m/s^2)

A. 9.00 ATM
B. 90.0 ATM
C. 100 ATM
D. 111 ATM

28. Water is flowing at 4.0 m/s in a circular pipe. If the diameter of the pipe decreases to 1/2 its former value, what is the velocity of the water downstream?

A. 1.0 m/s
B. 2.0 m/s
C. 8.0 m/s
D. 16 m/s

29. Which of the following is **NOT** an assumption usually made in the study of ideal fluid flow?

A. The fluid is nonviscous
B. The flow is steady.
C. The fluid is compressible.
D. The flow is irrotational.

30. A net force F accelerates a mass m with an acceleration a. If the same net force is applied to a mass 5m, what is the acceleration of this mass?

A. $5a$
B. $25a$
C. $a/5$
D. $a/25$

31. If the only forces acting on a 2.0-kg mass are $\vec{F}_1 = (3\hat{i} - 8\hat{j})$ N and $\vec{F}_2 = (5\hat{i} + 3\hat{j})$ N, what is the magnitude of the acceleration of the particle?

A. 1.5 m/s^2
B. 4.7 m/s^2
C. 6.5 m/s^2
D. 9.4 m/s^2

32. A soccer player kicks a soccer ball with a force of 1300 N. The soccer ball hits the player with a force

A. less than 1300 N
B. exactly 1300 N

C. more than 1300 N
D. equal to 0 N

33. A 4.00 kg ball is travelling at 5.00 m/s and strikes a wall. The ball bounces off the wall with a velocity of 4.0 m/s. The change in momentum of the ball is:

A. 8.0 kg m/s.
B. 13 kg m/s.
C. 26 kg m/s.
D. 36 kg m/s.

34. A ball collides with a second ball at rest. After the collision, the first ball comes to rest and the second ball moves off at the speed of the first ball. In this collision

A. total momentum and total kinetic energy are conserved and the masses are equal
B. total kinetic energy is not conserved
C. total momentum is conserved but total kinetic energy is not conserved
D. total momentum is not conserved but total kinetic energy is conserved

35. In the absence of a net force, a moving object will

A. slow down and eventually stop
B. stop immediately
C. move with constant velocity
D. turn right

36. If bullet of mass 20 g is fired from a gun of 5 kg with a velocity of 100 ms^{-1}, then recoil velocity of the gun will be

A. 0.4 ms^{-1}
B. 1,0 ms^{-1}

MYSTIC VENTURES

UNIVERSITY OF CAPE COAST

COLLEGE OF AGRICULTURE AND NATURAL SCIENCES

SCHOOL OF PHYSICAL SCIENCE

END OF FIRST SEMESTER EXAMINATION, 2016

PHY 101: GENERAL PHYSICS

DURATION: 2 HOURS

ANSWER ALL QUESTIONS.

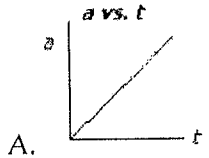
MYSTIC

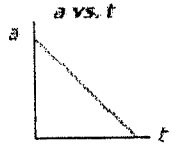
1. What is the product of 9.0 *ns* and 4.24 x 10^5 *Km/s* in SI units?
A. 3.8 *m*
B. 3.8 x 10^5 *m*
C. 3.8 x 10^8 *m*
D. 3.8 x 10^5 *Km/s*

2. The dimensions of acceleration are
A. L^3T
B. T^2L
C. LT^2
D. LT^{-2}

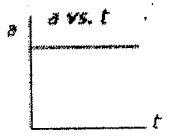
3. Which of the following is **NOT** an example of a vector quantity?
A. Force
B. Velocity
C. Speed
D. Acceleration

4. A ball is thrown vertically upward into the air. Which of the following acceleration-time graphs represents the ball's motion?

A. 

B. 

C. 

D. 

58. Some species of whales can dive to depths of one kilometer. What is the total pressure they experience at this depth? ($\rho_{seawater} = 1\,020\ kg/m^3$ and $1.01 \times 10^5\ N/m^2 = 1\ ATM$, g = 9.8 m/s^2)
A. 9.00 *ATM*
B. 90.0 *ATM*
C. 100 *ATM*
D. 111 *ATM*

59. Water is flowing at 4.0 m/s in a circular pipe. If the diameter of the pipe decreases to 1/2 its former value, what is the velocity of the water downstream?
A. 1.0 m/s
B. 2.0 m/s
C. 8.0 m/s
D. 16 m/s

60. Which of the following is **NOT** an assumption usually made in the study of ideal fluid flow?
A. The fluid is nonviscous
B. The flow is steady.
C. The fluid is compressible.
D. The flow is irrotational.

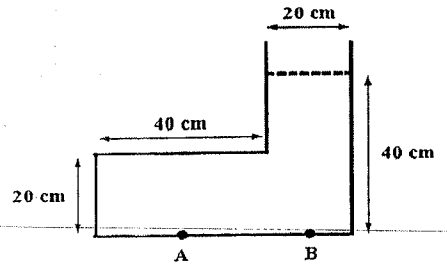
55. In a car lift used in a service station, compressed air exerts a force on a small piston that has a circular cross section and a radius of 5.00 *cm*. This pressure is transmitted by a liquid to a piston that has a radius of 15.0 *cm*. What force must the compressed air exert to lift a car weighing 13 300 *N*?

A. $4.10 \times 10^3\ N$
B. $8.60 \times 10^3\ N$
C. $1.96 \times 10^3 N$
D. $1.48 \times 10^3\ N$

56. A piece of wood is placed on top of the ice in a large bowl half full of ice. The bowl is then filled to the brim with water, with the wood block riding on top of the ice. As the ice melts,

A. the density of the water decreases.
B. the water level falls below the brim.
C. the water level rises and water spills out of the bowl.
D. the water level does not change.

57. The figure below shows a container filled with water to the height shown. When we compare the pressure at A to the pressure at B, we find that

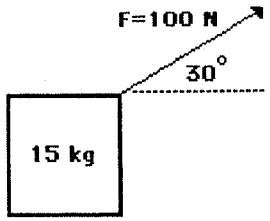


A. $p_A = \frac{1}{4} p_B$.
B. $p_A = \frac{1}{2} p_B$.
C. $p_A = p_B$.
D. $p_A = 2p_B$.

MYSTIC

VENTURES

5. A force of 100 N acts on the block at the angle shown in the diagram. The block moves a horizontal distance of 15.0 m. What is the work done by the applied force?



A. 413 N m
B. 342 N m
C. 234 Nm
D. 457 Nm

6. What is the potential gravitational energy of a 90kg object at the top of 80 m hill?

A. 20000 J
B. 2354 J
C. 4598 J
D. 70, 632 J

7. A box shaped piece of iron has a top area of $15cm^2$ and a height of 3cm. When a shearing force of 0.50N is applied to the upper surface, the upper surface displaces 4mm relative to the bottom surface. What is the shearing stress?

A. 333 Pa
B. 123 Pa
C. 250 Pa
D. 430 Pa

8. The force between the molecules of a liquid in contact with solid is

A. adhesive
B. cohesion
C. magnetic
D. repulsive

MYSTIC

9. Newton's first law of motion gives the concept of
A. Energy
B. Work
C. Momentum
D. Inertia

10. A 250 cm^3 flask of glass filled with seawater weighs 350 g. Calculate the density of seawater if empty flask weighs 100 g.
A. 5/7 g/cm^3
B. 7/5 g/cm^3
C. 2/5 g/cm^3
D. 1 g/cm^3

11. A reference frame attached to the earth:
A. is an inertial frame by definition
B. is an inertial frame because Newton's laws are applicable in the frame
C. cannot be an inertial frame because the earth is revolving round the sun
D. cannot be an inertial frame because the earth is rotating about its own axis.

12. Two forces of magnitude 12 N and 8 N are acting at a point. If the angle between the two forces is 60°, determine the magnitude of the resultant force?
A. 5.8 N
B. 28.6 N
C. 17.43 N
D. 45.0 N

13. A first year physics student is standing on a rotating merry-go-round without slipping. Which force provides the student's centripetal acceleration?
A. Centripetal force
B. Friction on the feet
C. Normal force
D. Centrifugal force

50. A uniform acceleration equal to steel ball of mass 40 g rolls down a frictionless plane, inclined at 30° to the horizontal, with a
A. 20.0 ms^{-2}
B. 10.0 ms^{-2}
C. 8.7 ms^{-2}
D. 5.0 ms^{-2}

51. Which of the following statements is NOT true for unit vectors?
A. $\hat{i}.\hat{i} = \hat{j}.\hat{j} = \hat{k}.\hat{k} = 1$
B. $\hat{i}.\hat{j} = \hat{i}.\hat{k} = j.\hat{k} = 0$
C. $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$
D. $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 1$

52. Find the resultant of three concurrent forces of 8 N, 12 N and 16 N acting at respective angles 30°, 60° and 90° relative to the +X-direction.
A. 43.0 N, 067°
B. 36.0 N, 180°
C. 30.4 N, 090°
D. 12.9 N, 000°

53. Given two vectors $\vec{M} = (3i - j + 2k)$ and $\vec{N} = (i + 2j - 5k)$. Find the angle between $\vec{M}$ and $\vec{N}$.
A. 26°
B. 38°
C. 61°
D. 64°

54. A boy pulls a toy with a force 35 N inclined at an angle Θ to the horizontal, If the effective force pulling the toy along the horizontal floor is 28 N, what is the value of Θ?
A. 30°
B. 37°
C. 39°
D. 63°

45. Which of the following is dimensionally a correct formula?
A. $v = ut + at$
B. $v + u = at$
C. $\frac{v}{u} = at$
D. $vt = u - a$

46. 1kWh is the unit of
A. Time
B. Power
C. Energy
D. Stress

47. If $M^aL^bT^c$ is the dimensional formula of momentum, and $M^xL^yT^z$ dimensional formula of energy find the value of $ax + by - cz$.
A. -1
B. 1
C. 3
D. 5

48. Which of the following is Unit of length?
A. Lunar Month
B. Kelvin
C. candela
D. Light year

49. The following are all vector quantities except
A. Displacement
B. Velocity
C. Distance
D. Force

MYSTIC VENTURES

14. A stationary skateboarder-1 with a mass of 50 kg pushes a stationary skateboarder-2 with a mass of 75 kg. After the push the skateboarder-2 moves with a velocity of 2 m/s to the right. What is the velocity of the skateboarder-2?
A. 3 m/s to the left
B. 2 m/s to the left
C. 1 m/s to the right
D. 2 m/s to the right

15. A body travels 45m in 20s. its average velocity is
A. 2.25 m/s
B. 15 m/s
C. 90 m/s
D. 135 m/s

16. How long does a bicycle with an acceleration of 0.8 m/s^2 take to go from 4 m/s to 12 m/s?
A. 6.4 s
B. 10 s
C. 15 s
D. 26 s

17. How long will a car starting from rest travel in 3 s at a constant acceleration of 4 m/s^2?
A. 6 m
B. 12 m
C. 18 m
D. 72 m

18. What is the distance covered in 5 s by a motorcycle accelerating at 1.2 m/s^2 if its initial velocity is taken to be 10 m/s?
A. 15 m
B. 25 m
C. 53 m
D. 65 m

19. Find the acceleration of a car that comes to a stop from a velocity of 60 m/s in a distance of 360m.
A. -5 m/s^2
B. 5 m/s^2
C. -10 m/s^2
D. 10 m/s^2

20. Ball A is dropped from a roof at the same time as ball B is thrown horizontally from the roof. Ball A
A. reaches the ground first
B. reaches the ground last
C. has the greater velocity when it reaches the ground
D. has the smaller velocity when it reaches the ground

21. What is the velocity of a stone after it has dropped from a cliff and has fallen 20 m? [Take g = 10 m/s^2]
A. 10 m/s
B. 20 m/s
C. 196 m/s
D. 392 m/s

22. What is the horizontal component of a balls velocity if it is thrown with initial velocity 4.0 m/s at an angle 40°?
A. 2.6 1m/s
B. 3.06 m/s
C. 3.40 m/s
D. 5.05 m/s

23. A ball is thrown vertically downward with velocity of 12 m/s. How far has the ball fallen in 1 s? [Take g=10m/s^2]
A. 6 m
B. 12 m
C. 17 m
D. 24 m

41. Accurately known digits and first doubtful digit are called
A. limitation of instruments
B. systematic error
C. frequent error
D. significant figures

42. One of physical quantities which is not a derived quantity
A. velocity
B. acceleration
C. force
D. length

43. Which of the following gives the correct units for force?
A. kgm/s^2
B. kgm^2/s^2
C. kg/ms^2
D. kgm^2/s

44. Prefix naming as *femto* has a factor of
A. 10^{-12}
B. 10^{12}
C. 10^{15}
D. 10^{-15}

37. Hooke's law is NOT applicable beyond
A. Elastic limit
B. Plastic limit
C. Fracture point
D. Ultimate strength

38. Dimensions of coefficient of viscosity is
A. $ML^{-1}T^{-1}$
B. $ML^{-1}T^{-2}$
C. $ML^{-2}T^{-1}$
D. $ML^{-1}T$

39. Zero digit between two significant figures is
A. significant
B. non significant
C. may not be significant
D. maybe significant

40. According to system international (SI units) amount of substance is measured in
A. moles
B. kilograms
C. meters
D. candelas

MYSTIC VENTURES

24. A javelin is thrown at an angle of 60° with initial velocity of 30 m/s. Find the range it takes 10 s to hit the ground.
A. 100 m
B. 120 m
C. 140 m
D. 150 m

25. A 500 g ball moves in a horizontal circle of radius 40 cm at 4 m/s. Its centripetal acceleration is
A. 10 m/s^2
B. 20 m/s^2
C. 40 m/s^2
D. 80 m/s^2

26. An object in uniform circular motion is being acted upon by a centripetal force F. If the radius of the objects path is to be doubled while its velocity remains the same, the new centripetal force must be
A. F/2
B. F
C. 2F
D. 4F

27. A ball is swung in a circle at the end of a string. A property of the ball that does not depend the length of the string is its
A. Angular velocity
B. Angular momentum
C. Linear velocity
D. Centripetal acceleration

28. What is the angle in radian swept by a body which moves through a quarter of a circular path?
A. $\pi/4$ rad
B. $\pi/2$ rad
C. π rad
D. 2π rad

29. The linear velocity of the rim of a wheel 0.8m m in diameter when the wheel turns at 3π rad/s.
A. 0.6π m/s
B. 1.2π m/s
C. 1.5π m/s
D. 2.4π m/s

30. Calculate the force needed to stretch a spring by an amount of 0.40 m if its stiffness constant is 150 N/m?
A. 375 N
B. 190 N
C. 110 N
D. 60 N

31. A spring with a stiffness constant 450.0 N/m is used to weigh some materials. Calculate the mass of the material that will stretch the spring by 10.5 cm from its normal length.
A. 0.482 kg
B. 4.82 kg
C. 48.2 kg
D. 482 kg

32. Which of these statements is/are true?
I. Tensile or longitudinal stress stretches a body in one direction only, and therefore causes an increase in the length of the body.
II. Compressional stress acts only in one direction and results in a reduction in the length of the body.
III. Bulk or hydrostatic stress acts uniformly and equal from all sides of the body's surface and causes the entire body's size to reduce.
A. I only
B. II only
C. III only
D. All of the above

33. Following are the basic types of stress except
A. Tensile stress
B. Compressive stress
C. Shear stress
D. Volumetric stress

34. Which of the following statements is/are true?
When tensile stress is applied axially on a circular rod its
I. diameter decreases
II. length increases
III. volume decreases
A. I Only
B. II Only
C. I and II only
D. All of the above

35. A rod, 120cm long and of diameter 3.0 cm is subjected to an axial pull of 18 kN. The stress in N/mm^2 is.
A. 22.57
B. 23.47
C. 24.57
D. 25.47

36. Tensile Strain is
A. Increase in length / original length
B. Decrease in length / original length
C. Change in volume / original volume
D. All of the above

VENTURE

MYSTIC

UNIVERSITY OF CAPE COAST, GHANA

SCHOOL OF PHYSICAL SCIENCES - DEPARTMENT OF PHYSICS

PHY 101: GENERAL PHYSICS 1 – QUIZ I (GROUP 2) Oct. 2017

Lecturer: Dr. Frederick Sam TIME: **40 minutes**

Answer all questions

1. Your friend wants to know his height using SI syatem. He is six feet tall, what is the equivalent height in meters?
   A. 1.83 m B. 6 m C. 0.83 m D. 1.38 m
2. Which of the following quantity has dimensions of $[ML^{-2}T^{-3}]$?
   A. Work B. Power C. Pressure D. Impulse
3. A student while doing an experiment finds that the velocity of an object varies with time as follows; $V = Xt^2 + Yt + Z$ If V and t are in SI units determine the units of the constants X, Y and Z
   A. $ms^{-3}, ms^{-2}, ms^{-1}$ B. $ms^{-2}, ms^{-2}, ms^{-1}$ C. $ms^{-3}, ms^{-1}, ms^{-1}$ D. $ms^{-3}, ms^{-2}, ms^{-3}$
4. The centripetal force (F) acting on a particle moving uniformly in a circle depends on the mass (m) of the particle, its velocity (V) and radius ® of the circle. Derive dimensionally formula for force (F).
   A. $F = \frac{kmr^2}{v}$ B. $F = \frac{kmv^2}{r}$ C. $F = \frac{krm^2}{v}$ D. $F = \frac{kmv^2}{r^2}$
5. Compute $7462/(98.1 - 92.4)$, and express the final answer in three significant figures.
   (a) 1.31×10^3 (b) 1.30×10^3 (c) 1.39×10^3 (d) none of the above
6. The mass of a solid cube is 856g, and each edge has a length of 5.35cm. Determine the density of the cube in basic SI unit.
   (a) $5.59 \times 10^3 kg/m^3$ (b) $5.59 \times 10^3 g/cm^3$ (c) 5.59kg/m^3 (d) 5.59g/cm^3
7. Which of the following statements are true? Circle all that apply. Answer: CD
   A. A vector is a large quantity and a scalar is a small quantity.
   B. A scalar quantity has a magnitude and a vector quantity does not.
   C. A vector quantity is described with a direction and a scalar is not.
   D. Scalar quantities are path dependent quantities and vector quantities are not.
8. In the Vector Addition Lab, Martin starts at the classroom door and walks: [2.0 m, West; 12.0 m, North; 31.0 m, West; 8.0 m, South; 3.0 m, East] Using either a scaled diagram or a calculator, determine the magnitude and direction of Martin's resulting displacement.
   A. 65 m, 72^0 B. 30.3 m, 172^0 C. 60 m, 33^0 D.10m, 272^0
9. A ball is thrown upwards, rises to its peak and eventually falls back to the original height. As the ball rises, its acceleration is upwards; as it falls, its acceleration is downwards. TRUE or FALSE

MYSTIC

$$\begin{vmatrix} 2i & 3i \\ 3j & -15j \\ 3k & 9k \end{vmatrix} = \begin{matrix} -1i \\ -16j \\ -6k \end{matrix}$$

Given two vectors $A = (2i - j + 3k)$ *and* $B = (i + 5j + 3k)$. Use this information to answer question 10 and 11.

10. Find |A-3B|
A. $13\sqrt{2}$ B. $-13\sqrt{2}$ C. $4\sqrt{11}$ D. $-4\sqrt{11}$

11. By making use of the Dot product find the angle between A and B.
A. 63^0 B. 116^0 C. 243^0 D. None of the above

12. A car travels at 20km/h for 1.5h at 30km/h for 2.0h, and at 40km/h for 0.75h. The car's average velocity is approximately.
A. 19km/h B. 21km/h C. 28km/h D. 30km/h

13. A ball is dropped from a window 46m above the ground. How long does it take the ball to reach the ground?
A. 0.5s B. 1.0s C. 2.0s D. 3.0s

14. The velocity of a train is reduced from 15m/s to 7m/s while travelling a distance of 90m. How much further will the train travel before coming to rest?
A. 25 m B. 35 m C. 45 m D. 55 m

15. Which of the following statements about position-time graphs are true? Circle all that apply. Answer: BCD
A. It cannot be used to represent the motion of objects with accelerated motion.
B. A straight, diagonal line is representative of an object with a constant velocity.
C. Accelerated objects are represented on position-time graphs by curved lines.
D. An object with a positive velocity will be represented on a position-time graph by a line with a positive slope and a negative velocity with a negative slope

16. A projectile is launched at an angle of 15 degrees above the horizontal and lands down range. For the same speed, what other projection angle would produce the same downrange distance?
A. 50 degrees B. 45 degrees C. 75 degrees D. 30 degrees

17. Two projectiles are fired at equal speeds but different angles. One is fired at angle of 30 degrees and the other at 60 degrees. The projectile to hit the ground first will be the one fired at (neglect air resistance) ____.
A. 30 degrees B. 60 degrees C. both hit at the same time

A projectile is fired with a horizontal velocity of 330 m/s from the top of a cliff 80 m high. Use this information to answer question 18, 19 and 20.

18. How long will it take to strike the level ground at base of the cliff?
A. 3.04 B. 4.04 C. 5.04 D. 6.04

19. How far from the front of the cliff will it strike?
A. 1230 m B. 1313 m C. 1330 m D. 1312 m

20. With what velocity will it strike the the level ground?
A. 232 m/s B. 352 m/s C. 342 m/s D. 332 m/s

BS/FSC/17/0033 VASTUS 16/11/2017.

MYSIC

UNIVERSITY OF CAPE COAST, GHANA

SCHOOL OF PHYSICAL SCIENCES - DEPARTMENT OF PHYSICS

PHY 101: GENERAL PHYSICS 1 – QUIZ II (GROUP 2) Nov. 2017

Lecturer: Dr. Frederick Sam TIME: 40 minutes

Answer all questions

1. According to Newton's third law, every force is accompanied by an equal and opposite *reaction* force. The reason that these forces do not cancel each other is ____
   A. the action force acts for a longer time period
   B. the two forces are not always in the same direction
   C. one of the two forces is greater than the other
   D. the two forces act upon different objects; only forces on the same object can balance each other

2. Helina weighs 360 N on Earth. What is Helina's mass on the moon where the force of gravity is approximately 1/6-th that of Earth's? _____ (take $g = 10\,ms^{-2}$)
   A. 3600 kg B. 60.00 kg C. 36.00 kg D. 1.67 kg

3. A 0.250-kg ball is thrown upwards with an initial velocity of 12.0 m/s at an angle of 30.0 degrees. Determine the acceleration of the ball when it has reached the peak of its trajectory. Assume negligible air resistance.
   A. 6 m/s^2 B. 36 m/s^2 C. 10 m/s^2 D. 9.8 m/s^2

4. Determine the applied force required to accelerate a 3.25-kg object rightward with a constant acceleration of 1.20 m/s^2 if the force of friction opposing the motion is 18.2 N. (Neglect air resistance.)
   A. 22.1 N B. 18.2 N C. 3.90 N D. 32.5 N

5. A 945-kg car traveling rightward at 22.6 m/s slams on the brakes and skids to a stop (with locked wheels). If the coefficient of friction between tires and road is 0.972, determine the distance required to stop.
   A. 92.61 m B. 36.8 m C. 26.8 m D. 36.61 m

6. A 50-kg platform diver hits the water below with a kinetic energy of 5000 Joules. The height (relative to the water) from which the diver dove was approximately ____ meters.
   A. 5 B. 10 C. 50 D. 100

7. An arrow is drawn back so that 50 Joules of potential energy is stored in the stretched bow and string. When released, the arrow will have a kinetic energy of ____ Joules.
   A. more than 50 B. 50 C. less than 50 D. between 50 and 100

MYSIC

8. A job is done slowly, and an identical job is done quickly. Both jobs require the same amount of ____, but different amounts of ____.
A. energy, work B. power, work C. work, energy D. work, power

9. Consider the head-on collision between a butterfly and the windshield of a high speed bus. Which of the following statements are true?
A. The magnitude of the force encountered by the bug is greater than that of the bus.
B. The magnitude of the impulse encountered by the bug is greater than that of the bus.
C. The magnitude of the momentum change encountered by the bug is greater than that of the bus.
D. The magnitude of the velocity change encountered by the bug is greater than that of the bus.

10. In a physics experiment, two equal-mass carts roll towards each other on a level, low-friction track. One cart rolls rightward at 2 m/s and the other cart rolls leftward at 1 m/s. After the carts collide, they couple (attach together) and roll together with a speed of _____. Ignore resistive forces.
A. 0.5 m/s B. 0.33 m/s C. 0.67 m/s D. None of the above

11. Which of the following statements is true of an object moving in a circle at a constant speed?
A. The object experiences a force which has a component directed parallel to the direction of motion.
B. Inertia causes objects to move in a circle.
C. There is an outwards force on the object as long as the net force is inwards.
D. The acceleration and the net force vector are directed perpendicular to each other.

12. What is the acceleration of a piece of dust on an old-fashioned record album, 15 cm from the center, if the record is spinning at 33.3 rpm?
A. 1.80 m/s^2 B. 1.52 m/s^2 C. 0.52 m/s^2 D. 18.0 m/s^2

13. A 5 N force is applied to a 3 kg ball to change its velocity from +9 m/s to +3 m/s. The impulse experienced by the ball is ____ N•s.
A. -2.5 B. – 10 C. -18 D. -45

14. What is the reading of a bathroom scale when a man of mass 72 kg stands on it while riding in an elevator accelerating upwards at 5.4 m/s^2?
A. 4.1×10^3 N B. 3.1×10^3 N C. 2.1×10^3 N D. 1.1×10^3 N

15. Which of the following statements are true about the motion of planets round the sun?
A. The force of gravity is the only force which acts upon the planets
B. The planets which are furthest from the sun have the greatest period.
C. The velocity vector is directed tangent to the elliptical path
D. All of the above

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11/20

MYSTIC

UNIVERSITY OF CAPE COAST, GHANA

SCHOOL OF PHYSICAL SCIENCES - DEPARTMENT OF PHYSICS

PHY 101: GENERAL PHYSICS 1 – QUIZ III (GROUP 2) **Dec. 2017**

Lecturer: Dr. Frederick Sam TIME: 45 minutes

1. According to the kinetic theory of gases, at absolute zero of temperature
   A. Water freezes
   B. Liquid helium freezes
   C. Molecular motion stops
   D. Liquid hydrogen freezes

2. Real gases show remarkable deviation from that of ideal gas behavior at
   A. High temperature and low pressure
   B. Low temperature and high pressure
   C. High temperature and high pressure
   D. Low temperature and low pressure

3. Estimate the mean free path of an air molecule of diameter 4.0×10^{-10} m at 273 K and one atmosphere (1.01×10^5 Pa). Given that $N_A = 6.02 \times 10^{23}$ and $R = 8.31 J/K$)
   A. $5.2 \times 10^{-8}\ m$ B. $6.2 \times 10^{-9}\ m$ C. $7.2 \times 10^{-10}\ m$ D. $8.2 \times 10^{-11}\ m$

4. The translational kinetic energy of gas molecule, for one mole of the gas is equal to
   A. $2/3(RT)$ B. $1/2(RT)$ C. $2/1(RT)$ D. $3/2(RT)$

5. What is the average velocity of the molecules in a sample of oxygen at 100 ^{0}C? The mass of an Oxygen molecule is $5.3 \times 10^{-26} kg$ [Boltzmann's constant $k = 1.38 \times 10^{-23} J/K$.]
   A. $6.4 \times 10^2 m/s$ B. $5.4 \times 10^2 m/s$ C. $4.4 \times 10^2 m/s$ D. $3.4 \times 10^2 m/s$

6. Find the mass of the ethyl alcohol molecule C_2H_6O given that the atomic masses of H, C and O are 1.008, 12.01 and 16.00 respectively. [$1\ amu = 1.66 \times 10^{-27} kg$.]
   A. $6.65 \times 10^{-26} kg$ B. $7.65 \times 10^{-26} kg$ C. $8.65 \times 10^{-26} kg$ D. $9.65 \times 10^{-26} kg$

7. How many molecules are in 1cm^3 of helium gas at 300K? ($P = 1.013 \times 10^5 Pa$ Avogadro's constant $N_A = 6.02 \times 10^{23}$ and $R = 8.31 J/K$)
   A. 6.4×10^{19} B. 5.4×10^{23} C. 2.4×10^{19} D. 2.4×10^{23}

8. Which of the following statements represent the first law of thermodynamics?
   A. Net heat input = change in internal energy + net heat output
   B. Net heat output = net work output + change in internal energy
   C. Net heat output = change in internal energy + net work output
   D. Net heat input = change in internal energy + net work output

9. A heat engine takes heat at one temperature and turns some of it into work and rejects the rest at a lower temperature. True or False

10. The rate of heat flow through a slab of material depends on the following except?
    A. The thickness of the slab
    B. The cross-section area of the slab

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C. The specific heat capacity
D. The temperature difference between the faces of the slab

11. How much work is done when a gas expands at constant pressure of 3 atmospheres from a volume of 2 to 5 liters? [1 atmosphere = $1.013 \times 10^5 Pa$]
A. 912 J B. 812 J C. 712 J D. 512 J

12. What is the Celsius equivalent of 80°F?
A. 36.7 °C B. 26.7 °C C. 16.7 °C D. 46.7 °C

13. Determine the amount of heat that will be added to 3 kg of water to raise its temperature from 20 to 80 ° C. [specific heat capacity of water is 4.185 KJ/(kg. ° C)]
A. 453 KJ B. 553KJ C. 653 KJ D. 753 KJ

14. The freezing point of water 0 °C. Its melting point is ----------
A. Slightly lower than 0 °C B. 0 °C C. Slightly higher than 0 °C D. 32 0 °C

15. The total mechanical energy of an object remains constant if the only forces doing work on the object are conservative forces. True or False

16. A job is done slowly, and an identical job is done quickly. Both jobs require the same amount of ____, but different amounts of ____.
A. energy, work B. power, work C. work, energy D. work, power

17. A 50-kg platform diver hits the water below with a kinetic energy of 5000 Joules. The height (relative to the water) from which the diver dove was approximately ____ meters.
A. 5 B. 10 C. 50 D. 100

18. A 60 kg box is pushed 12 m across a horizontal floor by a horizontal force of 200 N. If the coefficient of kinetic friction is 0.3, how much work went into overcoming friction?
A. 176 J B. 2112 J C. 1221 J D. 716 J

19. A hammer with 1.5 kg head is used to drive a nail into a wooden board. If the hammer head is moving at 3 m/s when it strikse the nail and the nail moves 10 mm into the board, find the average force the hammer head exert on the nail.
A. 450 kN B. 13.5 kN C. 4.5 kN D. 1.35 kN

20. An arrow is drawn back so that 50 Joules of potential energy is stored in the stretched bow and string. When released, the arrow will have a kinetic energy of ____ Joules.
A. more than 50 B. 50 C. less than 50 D. between 50 and 100

21. A man uses a horizontal force of 200 N to push a crate up an inclined plane 8 m long that is 20° above the horizontal. If Martin takes 12 sec to push the crate up the plane what is his power output?
A. 125 W B. 1504 W C. 166 W D. 1600 W

MYSTIC

UNIVERSITY OF CAPE COAST
COLLEGE OF AGRICULTURAL AND NATURAL SCIENCES
SCHOOL OF PHYSICAL SCIENCES
DEPARMENT OF PHYSICS

END OF FIRST SEMESTER EXAMINATION: 2015
PHY 101: GENERAL PHYSICS I (THEORY)

ANSWER ALL QUESTIONS

DURATION: 2 Hours

Choose from the letters A – D which bears the correct ANSWER

1. What is the product of (9.0 ns) and (4.24×10^5 km/s) in SI units?
   A. 30 km/s^2
   B. 250 m
   C. 380 km
   D. 3.8 m

VENTURES

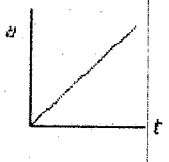
2. The SI unit equivalent of 7.6 g/cm^3 is
   A. 76 g/m^3
   B. 760 kg/cm^3
   C. 7.6 kg/m^3
   D. 7600 kg/m^3

3. If $x = a + bt + ct^2$, where x is in metres (m) and and t in seconds (s), which of the following respectively gives the dimensions of a, b and c?
   A. L, LT^{-1} and LT^{-2}
   B. L, LT and LT^2
   C. L^{-1}, $L^{-1}T$ and $L^{-1}T^2$
   D. LT^{-2}, LT^{-1} and L

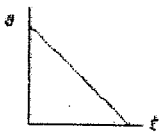
4. Name the physical quantity which is equal to the product of force and velocity.
   A. Work
   B. Energy
   C. Power
   D. Acceleration

Use the information below to answer questions 5, 6 and 7.
A ball is thrown vertically upward into the air at an initial velocity of 3 m/s.
5. Which of the following acceleration-time graphs represents its motion?

A



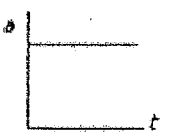
B



C



D



MYSTIC VENTURES

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6. What is the maximum height reached?
   A. 0.45 m
   B. 0.3 m
   C. 0.15 m
   D. 0.8 m

7. How long did it take to reach the maximum height?
   A. 0. 2 s
   B. 0.4 s
   C. 0.3 s
   D. 0.1 s

8. A body performing uniform circular motion has constant speed but variable velocity because
   A. it experiences a centripetal force.
   B. its direction of motion changes.
   C. its centripetal force and centrifugal forces balance each other
   D. it experiences a centrifugal force.

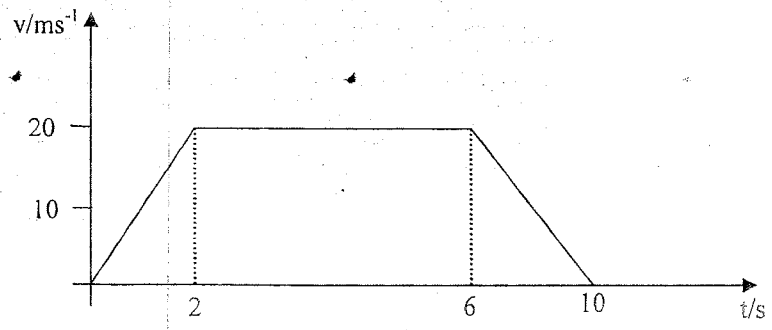
9. Which of the following equations represents the distance (s) travelled by a body falling freely under gravity? [The symbols g and t have their usual meanings].
   A. $s = gt^2$
   B. $s = 2gt^2$
   C. $s = gt^2/2$
   D. $s = 2gt$

10. A body starts from rest and travels distances of 100 m, 200 m and 300 m in successive equal time intervals of 10 s. Which of the following statements is/are true about the body?

    I. It has uniform acceleration.
    II. It has constant velocity.
    III. It has non-uniform velocity.

    A. I only
    B. II only
    C. I and III only
    D. I , II and III only

The graph below represents the motion of a body between two points. Use it to answer questions 11, 12 and 13.



11. The total distance covered will be:
    A. 1.4 m
    B. 14.0 m
    C. 140.0 m
    D. 1400.0 m

12. The acceleration of the body is
    A. 0.01 ms^{-2}
    B. 0.1 ms^{-2}
    C. 1.0 ms^{-2}
    D. 10 ms^{-2}

13. The deceleration of the body is
    A. 5.0 ms^{-2}
    B. 0.5 ms^{-2}
    C. 0.05 ms^{-2}
    D. 0.005 ms^{-2}

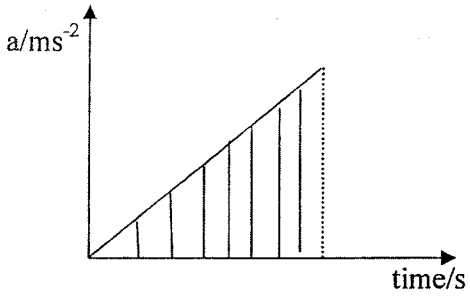
14. A plane inclined at an angle of 25° to the horizontal has a mass of 20 g resting on it. What is the coefficient of friction on the plane if the body just begins to slide?
    A. 1.565
    B. 0.466
    C. 2.336
    D. 6.44

15. Which of the following expressions is true for elastic collision? The coefficient of restitution *e* is
    A. $e = 0$
    B. $e < 0$
    C. $e = 1$
    D. $e > 1$

16. Each of the following quantities is a vector except
    A. distance
    B. force
    C. velocity
    D. displacement

17. A 0.25-kg ball attached to a string is rotating in a horizontal circle of radius 0.5 m. Assume the tension force is directed horizontally, and the ball revolves twice every second, what is the tension in the string?

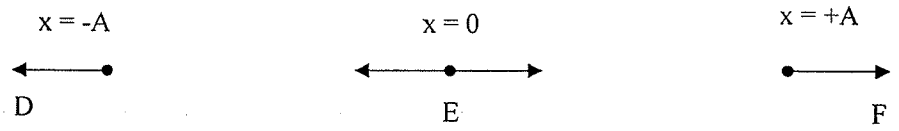
A. 2 N
B. 5 N
C. 7 N
D. 20 N



18. The area under an acceleration-time graph for a moving object, as depicted in the graph above, will give what information about the object?

A. total distance covered
B. velocity
C. acceleration
D. momentum

An object's motion is depicted below as performing a simple harmonic motion. Use this diagram to Answer questions 19 and 20.



19. The velocity of the object at position D will be

I. Maximum
II. Minimum
III. Zero

A. I only
B. II only
C. III only
D. II and III only

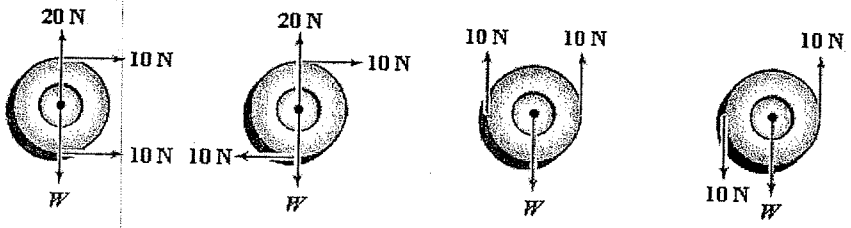
20. The acceleration of the object at position E will be
    I. Maximum
    II. Minimum
    III. Zero

    A. I only
    B. II only
    C. III only
    D. II and III only

21. Which of the following statements is/are true about a force acting on a body? It will
    I. get the body to start moving
    II. bring a moving body to rest
    III. alter the energy of a body

    A. I, II and III
    B. III only
    C. II only
    D. I only

22. The diagrams below show forces applied to a wheel that weighs 20 N. The symbol *W* stands for the weight. In which diagram(s) is (are) the wheel in equilibrium?



    A. P and S
    B. R
    C. Q and R
    D. P

23. Which of the following statements is/are true?
    I. The scalar product of the vectors $i.i = 1$
    II. The scalar product of the vectors $i.j = 0$
    III. The scalar product of the vectors $j.k = 1$

    A. I only
    B. II only
    C. III only
    D. I and II only

24. The property of matter which gives a measure of its resistance to changes in speed and direction of motion is:
    A. Mass
    B. Potential energy
    C. Kinetic energy
    D. Momentum

25. A body which is accelerating is observed to have constant speed. This means that the body is moving in.....
    A. an ellipse.
    B. a parabola.
    C. a straight line.
    D. a circle.

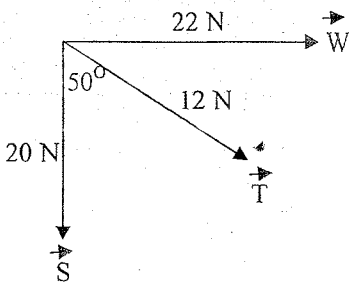
26. The frequency of a body moving with simple harmonic motion is doubled. If the amplitude remains the same, which one of the following is also doubled?
    A. the time period
    B. the maximum velocity
    C. the total energy
    D. the maximum acceleration

27. If vector **A** is perpendicular to vector **B**, what is the magnitude of **A+B**?
    A. $\sqrt{A^2 + B^2}$
    B. $A^2 + B^2$
    C. $A + B$
    D. $(A + B)^2$

28. A music compact disc does 45 revolutions in one minute. Calculate its angular velocity in radians per second.
    A. 1.50π
    B. 15.00π
    C. 45.00π
    D. 0.75π

29. A toy car of mass 2.0 kg is made to move in a circular track of radius 10.0 m. If the centripetal force acting is 980.0 N, calculate the angular velocity of the car.
    A. 7.0 $rads^{-1}$
    B. 96.0 $rads^{-1}$
    C. 49.0 $rads^{-1}$
    D. 20.0 $rads^{-1}$

30. For the figure below, the vector product $\vec{T} \times \vec{W}$ will be calculated as:



A. 242.42 N
B. 196.24 N
C. 185.52 N
D. 169.69 N

31. A wheel rotating about a fixed axis has an angular position given by $\Theta = 3.0 - 2.0t^3$, where Θ is measured in radians and t in seconds. What is the angular acceleration of the wheel at $t = 2.0$ s?
A. -1.0 $rads^{-2}$
B. -24 $rads^{-2}$
C. -2.0 rad s^{-2}
D. -4.0 $rads^{-2}$

32. Which of the following is not true concerning projectile motion?
A. Horizontal and vertical motions are independent of acceleration due to gravity.
B. Horizontal and vertical motions are independent of each other.
C. Time taken to cover horizontal range is twice that of maximum height.
D. Trajectory of the motion is parabolic.

33. A javelin thrower intends to reach the maximum horizontal range for his effort of throwing at 50 ms^{-1}. What should be his launch angle?
A. 35°
B. 60°
C. 45°
D. 50°

Use the information below to answer questions 34 and 35.
A stone is projected at an angle of 60° to the horizontal with a velocity of 30 m/s.

34. What is the total time of flight?
A. 1. 2 s
B. 2.3 s
C. 6.2 s
D. 5.2 s

35. What is the range?
A. 47 m
B. 70m
C. 68 m
D. 78 m

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36. The motion of a ball spinning on its axis is
    A. oscillatory
    B. rectilinear
    C. rotational
    D. circular

37. A body of mass 1000kg is released from a height of 10 m. With what energy does it strike the ground? Take g= $10m/s$
    A. 1.0×10^{5} J
    B. 1.0×10^{-5} J
    C. 10.0×10^{7} J
    D. 10.0×10^{-7} J

38. Which of these statements is/are true about collision between two or more bodies?
    I. During an elastic collision deformation is temporary
    II. During an inelastic collision momentum is not conserved
    III. In a system of colliding bodies the total kinetic energy before collision always equals the total kinetic energy after collision irrespective of the conditions of collision.

    A. I only
    B. II only
    C. III only
    D. I, II and III

39. An object A of mass 2 kg is moving with a velocity of 3 ms^{-1} and collides head-on with an object B of mass 1 kg moving in the opposite direction with a velocity of 4 ms^{-1}. Their common velocity, if after collision the objects stick together, will be calculated as
    A. 0.67 ms^{-1}
    B. 1.5 ms^{-1}
    C. 3.33 ms^{-1}
    D. 1 ms^{-1}

Use the following information to answer questions 40 and 41.
While waiting in her car at a spotlight, a 60 kg woman and her car are suddenly accelerated to a speed of 5 m/s as a result of a rear-end collision. If the time taken is 0.3 s:

40. The impulse on the woman can be calculated as:
    A. 1000 Ns
    B. 400 Ns
    C. 90 Ns
    D. 300 Ns

41. The average force exerted on her by the back of the seat of her car will be
    A. 300 N
    B. 400 N
    C. 1000 N
    D. 90 N

42. Tensile strain is defined as the ratio of:
    A. Change in volume to original volume
    B. Change in cross-sectional area to original cross-sectional area
    C. Change In length to original length
    D. Force to the cross-sectional area.

43. The slope of tensile stress to tensile strain curve, within elastic limits, gives
    A. Young's Modulus
    B. Bulk Modulus
    C. Shear Modulus
    D. Yield Point

44. Which of these statements is/are true about friction for a given surface?
    I. Coefficient of kinetic friction is always less than coefficient of static friction.
    II. Coefficient of kinetic friction is always greater than coefficient of static friction.
    III. The frictional force is independent of the area of contact of the given surface when the normal reaction is constant.

    A. I only
    B. II only
    C. III only
    D. I and III

45. What force is necessary to stretch a 2.0-mm diameter Cu wire ($Y = 11 \times 10^{10}$ N/m^2) by 1.0%?
    A. 2.1 kN
    B. 3.5 kN
    C. 6.9 kN
    D. 11.0 kN

46. Work done by a force is said to be positive if
    A. The angle between the applied force and the displacement is obtuse
    B. The applied force has a component in the direction of the displacement
    C. The applied force has a component in a direction opposite to the displacement
    D. The applied force is perpendicular to the displacement

47. What would be the final temperature of a mixture of 100 g of water at 90^{0}C and 600 g of water at 20^{0}C? SHC for water is 4186 J/kg^{0}C.
    A. 7.5^{0}C
    B. 15^{0}C
    C. 30^{0}C
    D. 60^{0}C

48. 2.00 L of a gas is known to be at a temperature of 27 ^{0}C and at 740.0 mmHg pressure. What is its volume at standard atmospheric pressure of 760 mmHg?
    A. 0.49 L
    B. 1.95 L
    C. 2.10 L
    D. 4.20 L

49. A large force acting for a very short period of time is:
    A. Retardation
    B. Momentum
    C. Collision
    D. Impulse

50. In order to establish a temperature scale the temperature at which it is expected that a particular physical event always takes place is:
    A. Thermometric temperature point
    B. Fixed point
    C. Triple point
    D. Temperature point

51. Which of the following is NOT a thermometric property?
    A. The colour of a gas
    B. Volume of a liquid
    C. The resistance of a wire
    D. The volume of a gas

52. The temperature of a quantity of an ideal gas is
    A. one measure of its ability to transfer thermal energy to another body.
    B. proportional to the average molecular kinetic energy of the molecules.
    C. proportional to the internal energy of the gas.
    D. correctly described by all the statements above.

53. The amount of heat required to raise the temperature of a body by $1^{o}C$ is called its
    A. specific heat
    B. melting energy
    C. latent heat
    D. boiling point

54. 45 ^{0}C converted to Fahrenheit degrees will be
    A. 7.22 ^{0}F
    B. 72.2 ^{0}F
    C. 11.3 ^{0}F
    D. 113 ^{0}F

55. Which statement below is ***NOT*** an assumption made in the molecular model of an ideal gas?
    A. The average separation between molecules is large compared with the dimensions of the molecules.
    B. The molecules undergo inelastic collisions with one another.
    C. The forces between molecules are short range.
    D. The molecules obey Newton's laws of motion.

56. If the temperature of a gas is increased, the kinetic energy would
    A. increase
    B. decrease
    C. remain same
    D. increase and decrease both

57. Which of these statements is/are true?
    I. An isotherm is a line on a P-V diagram that represents a constant-temperature process
    II. Boyle's law can be illustrated on an isotherm
    III. Charles law can be represented on an isotherm

    A. I only
    B. I and II only
    C. I and III only
    D. I, II and III

58. Calculate the quantity of heat conducted through 2 m^2 of a brick wall 12 cm thick in 1 hour if the temperature on one side is 8 ^{0}C and on the other side is 28 ^{0}C.
    (Thermal conductivity of brick = 0.13 $Wm^{-1}K^{-1}$)
    A. 156000 J
    B. 70000 J
    C. 20 J
    D. 24 J

MYSTIC

59. Which of these equations represent Boyle's law?
    I. $P_1V_1/T_1 = P_2V_2/T_2$
    II. $V_1/T_1=V_2/T_2$ = constant
    III. $P_1/P_2 = V_2/V_1$

    A. I only
    B. II only
    C. III only
    D. I, II, and III

VENTURE

60. Which of these statements is/are true about absolute zero temperature?
    I. At absolute zero, the kinetic energy of any given substance is zero
    II. -273.15 0C represents the temperature value at absolute zero
    III. Absolute zero is the lowest possible attainable temperature

    A. I only
    B. B. II only
    C. C. III only
    D. D. I, II, and III

Fundamental Constants

Quantity	Symbol	Value
acceleration due to gravity	g	$9.8\ ms^{-2}$
Avogadro's number	N_A	$6.022\ 141\ 99 \times 10^{23}\ mol^{-1}$
Boltzmann's constant	k	$1.380\ 6503 \times 10^{-23}\ J/K$
Electron charge magnitude	e	$1.602\ 176\ 462 \times 10^{-19}\ C$
Permeability of free space	μ_o	$4\pi \times 10^{-7}\ T.m/A$
Permittivity of free space	ε_o	$8.854 \times 10^{-12}\ C^2/(N.m^2)$
Plank's constant	h	$6.626 \times 10^{-34}\ J.s$
Mass of electron	m_e	$9.109 \times 10^{-31}\ J.s$
Mass of neutron	m_n	$1.674 \times 10^{-27}\ kg$
Mass of proton	m_p	$1.672 \times 10^{-27}\ kg$
Speed of light in vacuum	c	$2.997 \times 10^{8}\ m/s$
Universal gravitational constant	G	$6.673 \times 10^{-11}\ N.m^2/kg^2$
Universal gas constant	R	$8.314\ J/(mol.K)$
Coulomb's constant	κ	$8.99 \times 10^{9}\ N.m^2/C^2$
Electron volt	$1\ eV$	$1.6 \times 10^{-19}\ J$

UNIVERSITY OF CAPE COAST
DEPARTMENT OF PHYSICS
END-OF-FIRST SEMESTER EXAMINATIONS (2013/2014 ACADEMIC YEAR)

PHY 101: GENERAL PHYSICS I

Answer All Questions. *Time: 2 Hours*

(1) Each of the following is a vector quantity except
A. Weight
B. Velocity
C. Volume
D. Acceleration

(2) Which of the following units does **not** measure a scalar quantity?
A. Degree Celsius
B. Kilogram
C. Newton
D. Watt

Use the following information to answer questions (3) and (4).
Given the vectors $\mathbf{A} = 2\boldsymbol{i} + 6\boldsymbol{j} - \boldsymbol{k}$ and $\mathbf{B} = \boldsymbol{i} - 2\boldsymbol{j} + 2\boldsymbol{k}$,

(3) Find $2\mathbf{A} + \mathbf{B}$.
A. $3\boldsymbol{i} + 4\boldsymbol{j} - \mathbf{k}$
B. $\mathbf{5i - 10\,j + k}$
C. $\mathbf{3i + 8j + 3k}$
D. $\mathbf{5i + 10j + 0k}$

(4) Find $|\mathbf{B+A}|$.
A. 4.7
B. 5.1
C. 7.2
D. 8.0

(5) Two forces of magnitude 5N and 6N act on a particle at right angles to each other. Find the magnitude of their resultant.

A. 5.18 N
B. 7.18 N
C. 7.60 N
D. 7.81 N

MUSTAPHA YAHAYA

(6) Find $C \times D$, if C=3i - 2j and D=5i + 6j

A. 26k
B. 26j
C. 28k
D. 28j

(7) If a man walks 30.0 km due north and then walks 20.0 km due east. Compute the magnitude of the man's displacement from his starting position.

A. 10.0 km
B. 25.2 km
C. 36.1 km
D. 40.1 km

(8) A force of 40 N applied at an angle of 60° to the horizontal is used to drag a load along the floor of a ware house through a distance of 25 m. What is the work done in moving the load?

A. 25 J
B. 50 J
C. 250 J
D. 500 J

(9) Find the angle between two vectors A= 3i-4j and B=-2i+3k

A. 60.5°
B. 100.5°
C. 109.5°
D. 120.5°

(10) The force F acting on a particle has its line of action (direction) lying in the XY plane and is inclined at an angle θ with the x-axis. What is the z-component of the force F?

A. Zero
B. Fcos θ
C. F sin θ
D. F

(11) Find the magnitude of D if vector C=5i+ 2j-7k and D =2C.

A. 8.8
B. 17.7
C. 26.6
D. 78.0

(12) According to the kinetic theory of gases at absolute zero temperature

A. Water freezes
B. Liquid helium freezes
C. Molecular motion stops
D. Water attains minimum volume

(13) Which of the following properties of gas molecule is same for all ideal gases at a particular temperature?

A. Mass
B. Velocity
C. Momentum
D. Kinetic energy

(14) When a gas is heated from a temperature of 300 K to 600 K. The mean translational kinetic energy is changed by a factor of

A. 0.5
B. 2.0
C. 3.0
D. 4.0

(15) Which of the following statements about gas molecules is NOT correct? The gas molecules

A. are in a constant state of motion
B. have different speeds
C. have temperature which is a measure of the average kinetic energy
D. have no intermolecular attractive force

(16) What type of motion is generally performed by the molecules of a gas in a container?

A. Circular
B. Brownian
C. Rectilinear
D. Rotational

(17) The internal energy of one mole of an ideal gas depends on its

A. Volume
B. Temperature
C. Nature
D. Density

MYSTIC

MYSTIC VENTURES

(18) Lead melts at 626°F. The Celsius equivalent of this temperature is

A. 316°C
B. 330 °C
C. 366°C
D. 1069°C

(19). When a liquid evaporates.....

A. it gives off heat
B. it absorbs heat
C. its temperature drops
D. its temperature rises

(20). What will be the new temperature when 20kJ of heat is removed from 1.2kg of ice originally at -15°C

A. −18°C
B. −23°C
C. −26°C
D. −35°C

(21). The energy that must be added (or removed) to cause the solid – liquid transition in 1kg of a given material is called

A. Latent heat of transition
B. Latent heat of vaporization
C. Latent heat of fusion
D. Latent heat capacity

(22). Five kilograms of water at 40°C is poured on a large block of ice at 0°C. How much ice melts?

A. 2.5 kg
B. 25 kg
C. 3.6kg
D. 36kg

(23). Heat transfer by conduction occurs

A. Only in liquids
B. only in solids
C. Only in liquids and solids
D. in solids, liquids, and gases

(24). Which of the following is/are true about the rate of heat flow through a slab of a solid material?. The rate of heat flow depends on.....

I. the thickness of the slab
II. the area of the slab
III. the specific heat capacity of the material
IV. the temperature difference between the faces of the slab

A. I only
B. I and II only
C. I and III only
D. I, II and IV only

(25). The thermal conductivity of ice is 5.2×10^{-4} kcal / (m·s·°C). At what rate is heat lost by the water in a 6-m by 10-m outdoor swimming pool covered by a layer of ice 1 cm thick if the water is at a temperature of 0°C and the surrounding air is at a temperature of −10°C?

A. 27.1 kcal/s
B. 28.1 kcal/s
C. 30.1 kcal/s
D. 31.2 kcal/s

(26). An aluminium pot whose volume is 1000 cm^3 at 20°C has a volume of 1006 cm^3. The coefficient of linear expansion of aluminium is

A. 2.5×10^{-5}°C
B. 6.0×10^{-5}°C
C. 7.5×10^{-5}°C
D. 2.25×10^{-4}°C

(27). The pressure and absolute temperature of a gas sample whose volume is fixed are related by which one or more of the following formulas?

A. $\frac{P_1}{T_2} = \frac{P_2}{T_1}$
B. $\frac{P_1}{T_1} = \frac{P_2}{T_2}$
C. $\frac{P_1}{p_2} = \frac{T_2}{T_1}$
D. $\frac{P_1}{T_1} = \frac{T_2}{P_2}$

MYSTIC

(28). An object's temperature is raised by 100°C. The resulting increase in its absolute temperature is

A. 32K
B. 180K
C. 100K
D. 373K

(29). A sample of oxygen whose volume at 0°C and 200 kPa of pressure is 4.00 L is compressed to 1.00 L and its temperature is raised to 273°C. The pressure of the gas is now

A. 100 kPa
B. 400 kPa
C. 800 kPa
D. 1600 kPa

(30). Which of the following is NOT a thermometric property?

A. The expansion of a gas when heated
B. The contraction of a liquid when cooled
C. The melting of a solid when heated
D. The increase in the resistance of a conductor when heated

(31) Under which of the following conditions is the magnitude of the average velocity of a particle moving in one dimension smaller than the average speed over some time interval?

A. A particle moves in the $+x$ direction without reversing.
B. A particle moves in the $-x$ direction without reversing.
C. A particle moves in the $+x$ direction and then reverses the direction of its motion.
D. There are no conditions for which this is true.

(32) Which one of the following statements is true?

A. If a car is traveling eastward, its acceleration must be eastward.
B. If a car is slowing down, its acceleration must be negative.
C. A particle with constant acceleration can never stop and stay stopped.
D. Acceleration is not constant, if a car travels with constant velocity

(33). The velocity of a particle moving along the *x-axis* varies according to the expression

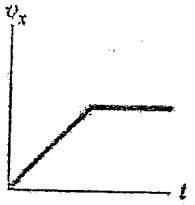
$$v_x = 40 - 5t^2,$$

where v_x is in meters per second and t is in seconds. What is the average acceleration in the time interval $t = 0$ to $t = 2.0$ s.

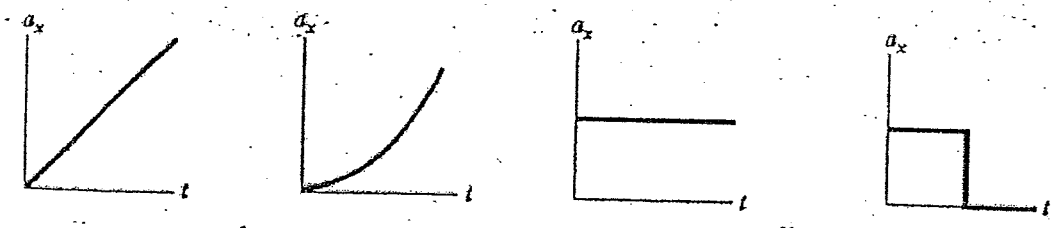
A. -10 m/s
B. +10 m/s
C. -10 m/s^2
D. +10 m/s^2

MYSTIC

(34) Matching the v_x–t graph below with a_x–t graphs, which of the graphs best describes the motion.



(A) (B) (C) (D)



(35) A jet lands on an aircraft carrier at a speed of 63 m/s. What is its acceleration (assumed constant) if it stops in 2.0 s due to an arresting cable that snags the jet and brings it to a stop?

A. +32 m/s^2
B. - 32 m/s
C. +32 m/s
D. - 32 m/s^2

(36) As a projectile thrown upward moves in its parabolic path, at what point along its path are the velocity and acceleration vectors for the projectile parallel to each other?

A. Nowhere
B. The highest point
C. The launch point
D. The landed point

(37) A baseball is thrown from the outfield toward the catcher. When the ball reaches its highest point, which statement is true?

My Tur Venture

A. Its velocity and its acceleration are both zero.
B. Its velocity is not zero, but its acceleration is zero.
C. Its velocity is perpendicular to its acceleration.
D. Its acceleration depends on the angle at which the ball was thrown.

Use the information below to answer questions (38) and (39):
A particle moves in a circular path of radius r with speed v. It then increases its speed to $2v$ while traveling along the same circular path.

(38) The centripetal acceleration of the particle has changed by what factor?
A. 0.25
B. 0.5
C. 2.0
D. 4.0

(39) By what factor has the period of revolution of the particle changed?
A. 0.25
B. 0.5
C. 2.0
D. 4.0

(40) A particle moves along a path, and its speed increases with time. In which case are its acceleration and velocity vectors perpendicular everywhere along the path?

A. When the path is circular
B. When the path is straight
C. When the path is a parabola
D. Never

(41) A certain light truck can go around a curve having a radius of 150 m with a maximum speed of 32.0 m/s. To have the same acceleration, at what maximum speed can it go around a curve having a radius of 75.0 m?

A. 64 m/s
B. 45 m/s
C. 32 m/s
D. 23 m/s

(42) A ball is connected to a light spring suspended vertically. When pulled downward from its equilibrium position and released, the ball oscillates up and down. In the system consisting of *the ball, the spring, and the Earth,* what forms of energy are there during the motion?
A. Kinetic and elastic potential
B. Kinetic and gravitational potential
C. Kinetic, elastic potential and gravitational potential
D. Elastic potential and gravitational potential

(43) Find the acceleration due to gravity in a space laboratory if a swinging pendulum of length 50 cm, takes 3.14 seconds to complete on oscillation.

A. 10.0 ms^{-2}
B. 9.8 ms^{-2}
C. 2.0 ms^{-2}
D. 16.0 ms^{-2}

(44) The displacement of a particle can be described by the equation $At^2 + {}^{B}/_{t}$. The respective dimensions of the constants A and B are

A. ML^2 and ML
B. LT^2 and LT^{-1}
C. MT and MT^{-2}
D. LT^{-2} and LT

MYSTIC

(45) Which of the following is a derived quantity?

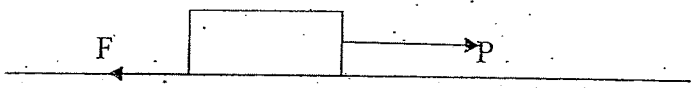
A. Speed
B. Mass
C. Time
D. Length

VELOCITY

(46) The speed limit when driving through a hospital zone is 40 km/h. This is equivalent to

A. 11.1 ms^{-1}
B. 90.0 ms^{-1}
C. 144.8 ms^{-1}
D. 500.0 ms^{-1}

(47) The figure below shows a block of mass 20 kg, being pulled along a horizontal floor with a force P = 50 N against a frictional force F = 10 N.



Find the acceleration of the block.

A. 2 ms^{-2}
B. 3 ms^{-2}
C. 4 ms^{-2}
D. 6 ms^{-2}

(48) What is the momentum of an asteroid that has a mass of 1.35×10^{12} kg and a velocity of 2.55×10^{4} ms^{-1}?

A. 1.89×10^{-8} $kgms^{-1}$
B. 5.29×10^{7} $kgms^{-1}$
C. 3,44 $\times 10^{16}$ $kgms^{-1}$
D. 8.78×10^{20} $kgms^{-1}$

(49) The average human body temperature on the Fahrenheit scale is

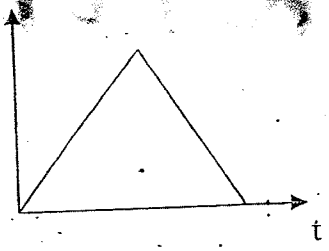
A. 102 °
B. 99 °
C. 67 °
D. 37 °

(50) The tendency of a body to remain at rest or in its state of uniform motion is known as

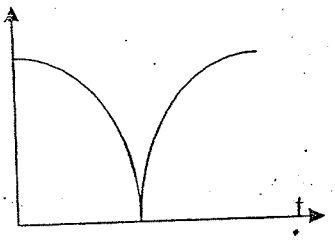
A. friction
B. gravity
C. momentum
D. inertia

(51) Which of the following correctly represents the displacement-time graph of a tennis ball bounced once?

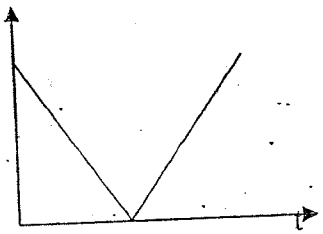
A. s



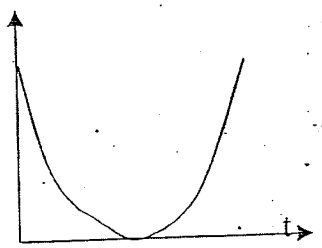
B. s



C. s



D. s



(52) The path taken by a projectile is known as its

A. orbit
B. range
C. parabola
D. trajectory

(53) A ball is kicked with a horizontal velocity of 25 ms^{-1} from the top of a cliff of height 45 m. Calculate the distance from the foot of the cliff where the ball lands. (Take g = 10 ms^{-2}).

A. 75 m
B. 70 m
C. 25 m
D. 20 m

(54) The displacement of a body performing simple harmonic motion is given by the equation $y = 8 \sin 200t$, where y is in cm and t is in seconds. The maximum velocity of the body is

A. 2.0 ms^{-1}
B. 2.5 ms^{-1}
C. 8.0 ms^{-1}
D. 16.0 ms^{-1}

(55) At what angle of inclination will a metal block begin to slip down a plane if the coefficient of friction of the surface is 0.4?

A. 66°
B. 45°
C. 24°
D. 22°

(56) What will be the new temperature when 20kJ of heat is removed from 1.2kg of ice originally at -15°C.

A. −18°C
B. −23°C
C. −26°C
D. −35°C

(57) The quantity of energy that must be added (or removed) to cause the solid – liquid transition in 1kg of a given material is called

A. Specific heat of transition
B. Specific latent heat of vaporization
C. Specific latent heat of fusion
D. Specific heat capacity

(58) Which of the following is/are true about the rate of heat flow through a slab of a solid material?. The rate of heat flow depends on.....

V. the thickness of the slab
VI. the area of the slab
VII. the specific heat capacity of the material
VIII. the temperature difference between the faces of the slab

A. I only
B. B I and II only
C. C. I and III only
D. D. I, II and IV only

59) The thermal conductivity of ice is 5.2×10^{-4} kcal / (m·s·°C). At what rate is heat lost by the water in a 6-m by 10-m outdoor swimming pool covered by a layer of ice 1 cm thick if the water is at a temperature of 0°C and the surrounding air is at a temperature of −10°C?

A. 27.1 kcal/s
B. 28.1 kcal/s
C. 30.1 kcal/s
D. 31.2 kcal/s

60) Calculate the specific heat capacity of copper given that 204.75 J of energy raises the temperature of 15g of copper from 25°C to 60°C.

A. 7166.3 $Jkg^{-1}K^{-1}$

B. 3071.3 $Jkg^{-1}K^{-1}$

C. 107.5 $Jkg^{-1}K^{-1}$

D. 390.0 $Jkg^{-1}K^{-1}$

UNIVERSITY OF CAPE COAST
COLLEGE OF AGRICULTURAL AND NATURAL SCIENCES
SCHOOL OF PHYSICAL SCIENCES
DEPARTMENT OF PHYSICS

END-OF-FIRST SEMESTER EXAMINATION – 2014/2015 ACADEMIC YEAR

PHY 101: GENERAL PHYSICS I (THEORY) *Time Allowed: 2 Hours*

INSTRUCTIONS: Answer all questions by writing the letter (A – D) corresponding to the correct option for each question in the answer booklet.

(Unless otherwise stated, take acceleration due to gravity $g = 10\ ms^{-1}$)

1. Which of the following is **NOT** a derived unit?

A. candela
B. joule
C. newton
D. watt

2. Which of the following quantities has the same dimensions as kinetic energy, $\frac{1}{2}mv^2$?

Note: $[a] = [g] = LT^{-2}$; $[h] = L$ and $[v] = LT^{-1}$.

A. *Ma*
B. *Mvx*
C. *Mvt*
D. *Mgh*

3. If $\vec{A} = 12\hat{i} - 16\hat{j}$ and $\vec{B} = -24\hat{i} + 10\hat{j}$, what is the direction of the vector $\vec{C} = 2\vec{A} - \vec{B}$?

A. -49°
B. -41°
C. -90°
D. +49°

4. Which of the following physical quantities has the dimensions MLT^{-1}?

A. Force
B. Speed
C. Density
D. Momentum

5. Which of the following numbers is correctly expressed in standard scientific notation?

A. 860×10^{7}
B. 64,829,000
C. 0.0854
D. 8.25×10^{-23}

6. What is the product of (9.0 ns) and (4.24×10^{5} km/s) in SI units?

A. 3.8 m
B. 250 ms^{-2}
C. 380 km
D. 30 kms^{-2}

7. The SI unit equivalent of 7.6 gcm^{-3} is

A. 76 gm^{-3}
B. 760 $kgcm^{-3}$
C. 7.6 kgm^{-3}
D. 7600 kgm^{-3}

8. Two balls of the same diameter, made of iron and cotton, are dropped from a height 'h' in air. Assuming air resistance is negligible, which of the following statements is true about their free fall?

A. The cotton ball will reach the ground first
B. The iron ball will reach the ground first
C. The two balls will reach the ground at the same time
D. The two balls will hit the ground with the same force

9. The earth's gravitational pull on objects

A. is the same at all places on the surface of the earth
B. is greater at the equator of the earth than at the poles
C. decreases with altitude
D. increases with altitude

10. A 2.0-kg block slides on a frictionless plane inclined at 25° to the horizontal. A force of 4.6 N acting parallel and up the incline is applied to the block. What is the acceleration of the block?

A. 1.8 ms^{-2} up the incline
B. 2.3 ms^{-2} up the incline
C. 6.6 ms^{-2} down the incline
D. 1.8 ms^{-2} down the incline

11. A man in a lift moving upwards with a constant velocity feels that his apparent weight

A. is zero
B. has increased
C. has decreased
D. is constant

12. Two forces acting on a body are said to be in equilibrium only if they are

A. equal in magnitude and opposite in direction
B. equal in magnitude and perpendicular to each other
C. not equal but act at right angle to each other
D. equal in magnitude and act along the same direction

13. Which of the following statements is/are true?
I. The scalar product of the vectors $i.i = 1$
II. The scalar product of the vectors $i.j = 0$
III. The scalar product of the vectors $j.k = 1$

A. I only
B. II only
C. III only
D. I and II only

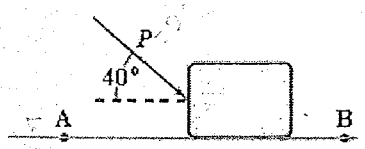
14. Three vectors are given by A = i - 2j, B = - 2i + j, C = -3i - 5j.
What is the *direction of the sum* of these vectors?

A. 56.3°
B. -56.3°
C. 33.7°
D. -33.7°

15. How much work is done by a person lifting a 2.0-kg object from the bottom of a well at a constant speed of 2.0 m/s for 5.0 s?

A. 0.22 kJ
B. 0.20 kJ
C. 0.24 kJ
D. 0.27 kJ

16. A block slides on a rough horizontal surface from point A to point B. A force (magnitude $P = 2.0$ N) acts on the block between A and B, as shown. Points A and B are 1.5 m apart. If the kinetic energies of the block at A and B are 5.0 J and 4.0 J, respectively, how much work is done on the block by the force of friction as the block moves from A to B?



A. -3.3 J
B. +1.3 J
C. +3.3 J
D. -1.3 J

17. An automobile traveling along a straight road increases its speed from 30.0 m/s to 50.0 m/s in a distance of 180 m. If the acceleration is constant, how much time elapses while the auto moves this distance?

A. 6.00 s
B. 4.50 s
C. 3.60 s
D. 4.00 s

18. John and Linda are arguing about the definition of density. John says the density of an object is proportional to its mass. Linda says the object's mass is proportional to its density and to its volume. Which one, if either, is correct?

A. They are both wrong.
B. John is correct, but Linda is wrong.
C. John is wrong, but Linda is correct.
D. They are both correct.

19. In 2.0 s, a particle moving with constant acceleration along the x axis goes from $x = 10$ m to $x = 50$ m. The velocity at the end of this time interval is 10 m/s. What is the acceleration of the particle?

a. +15 ms^{-2}
b. +20 ms^{-2}
c. -20 ms^{-2}
d. -10 ms^{-2}

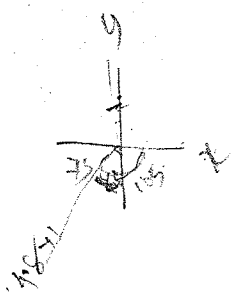
20. Deduce the *x*- and *y*- *components* of a vector P, if its direction is 105° counterclockwise from the x-axis and its magnitude is 4.8 N.

A. x = 4.8 *cos* 15°, y = 4.8 *sin* 15°
B. x = 4.8 *sin* 15°, y = 4.8 *cos* 15°
C. x = 4.8 *cos* 75°, y = 4.8 *sin* 75°
D. x = 4.8 *sin* 75°, y = 4.8 *cos* 75°

Use this preamble to answer the next two questions.
A man walks in the following pattern: 3.1 km *N*, 2.4 km *W* and 5.2 km *S*.

21. In what direction will a bird fly if it is to arrive at the same final destination within the shortest possible time?

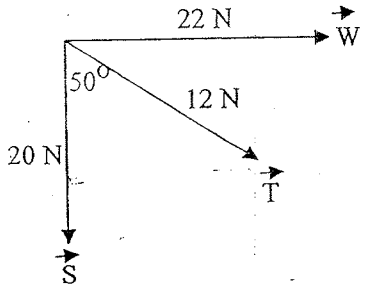
A. 41° S of W
B. 49° S of W
C. 37° N of W
D. 52° N of W



22. Calculate the distance covered by the bird.

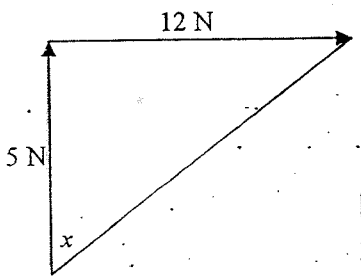
A. 2.11 km
B. 5.50 km
C. 10.70 km
D. 10.17 km

23. Calculate the vector product $T \times W$ (as shown in the figure below).



A. 242.42 N
B. 196.24 N
C. 185.52 N
D. 169.69 N

Use this preamble to answer the next two questions.
Consider two forces of magnitude 5 N and 12 N acting as shown below.



24. Calculate the resultant force.

A. 7 N
B. 13 N
C. 17 N
D. 60 N

25. What is the value of the angle x, the direction of the resultant force?.

A. 22.6°
B. 67.4°
C. 112.6°
D. 157.4°

26. A body rolls down a frictionless plane inclined at an angle θ to the horizontal. It falls with an acceleration equal to

A. gtanθ
B. g/θ
C. gcosθ
D. gsinθ

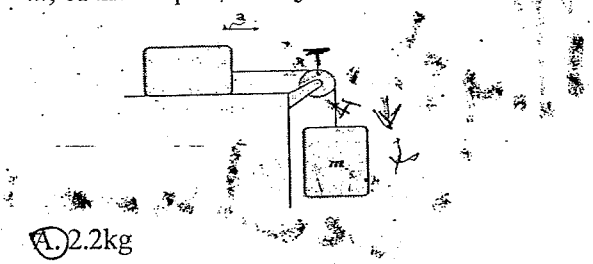
27. The vertical component of the displacement of a projectile with the initial velocity u, inclined at an angle θ to the horizontal, can be expressed as

A. $u^2 \sin\theta + gt$
B. $ut.\sin\theta - \frac{1}{2}gt^2$
C. ut.cosθ
D. ½ug.cos2θ

28. An athlete throws a javelin with a velocity of $29.4 ms^{-1}$ at an angle of 41° relative to the horizontal ground. Neglecting air resistance, calculate how far the javelin lands from the launch position.

A. 133.1 m
B. 66.5 m
C. 70.4 m
D. 22.2 m

29. If the tension, *T*, is 15 N and the magnitude of the acceleration, *a*, is 3.0 m/s^2, what is the mass, *m*, of the suspended object? Assume that all surfaces and the pulley are frictionless.



A. 2.2kg
B. 5.0kg
C. 3.3kg
D. 3.0kg

30. A javelin thrower intends to reach the maximum horizontal range for his effort of throwing at 50 ms^{-1}. What should be his launch angle?
A. 35°
B. 60°
C. 45°
D. 50°

UNIVERSITY OF CAPE COAST
DEPARTMENT OF PHYSICS
END-OF-FIRST SEMESTER EXAMINATIONS (NOVEMBER 2012)

PHY 101: GENERAL PHYSICS 1 (THEORY) Time Allowed: 2 hours

ANSWER ALL QUESTIONS. Choose the correct answer for each question from the options lettered A-D and write your answers in the answer booklet. Any rough work should be done on the unused pages in the answer booklet.

Physical Constants:

Acceleration due to gravity, $g = 9.8\ ms^{-2}$
Atmospheric Pressure $= 1.01 \times 10^5$ Pa
Density of water, $\rho_w = 1.00 \times 10^3\ kgm^{-3}$

1. Which of the following is a derived unit?

A. Mole
B. kilogram
C. Seconds
D. Newton

2. The density of a solid is 20 kg/m^3. The equivalence of this in g/cm^3 is

A. 2×10^4
B. 2×10^2
C. 2×10^{-2}
D. 2×10^{-3}

3. What is the product of (9.0 μs) and (4.5×10^5 km/s) in SI units?

A. 4.05 km
B. 40.5 m
C. 40.5 km
D. 4.05 m

4. If $x = a + bt + ct^2$, where x and a are in metres (m), b is in metre per second (m/s). Which of the following is the dimension of c?

A. LT^{-1}
B. L^2T
C. LT^{-2}
D. $L^{-1}T$

5. Which of the following is dimensionally equivalent to pressure?

A. Force
B. Stress
C. Strain
D. Inertia

6. Which of the following quantities has both magnitude and direction?

A. Temperature
B. Time
C. Energy
D. Velocity

7. A body of mass 5.0 kg is pulled along a frictionless horizontal ground by means of a force of 40N acting at an angle of 60° to the horizontal. The effective force pulling the body along the ground is

A. 15.4N
B. 20.0N
C. 34.6N
D. 40.0N

8. Two forces 3N and 4N act on a 20 kg mass. Calculate the magnitude of the resultant force on the mass if the angle between the two forces is 60^0.

A. 37.00N
B. 42.39N
C. 45.79N
D. 54.23N

9. Find the angle between the vectors $A = 2i + 2j$ and $B = 4i + 4j$.

A. 0^0
B. 35^0
C. 75^0
D. 90^0

10. Find the magnitude of Q – R if $Q = 4i - 2j + 3k$ and $R = 5i + 3j - 2k$.

A. 3.23
B. 5.61
C. 7.14
D. 9.51

11. A helicopter blade has as angular speed of $\omega = 6.50$rev/s and an angular acceleration $\alpha = 1.30$rev/s^2. If the radius of the helicopter blade is 3.0m, find the magnitude of the tangential speed.

A. 122.5 ms^{-1}
B. 224.5 ms^{-1}
C. 254.6 ms^{-1}
D. 300.2 ms^{-1}

12. A driver completes $3\frac{1}{2}$ somersaults in 1.7s. What is the average angular speed (in rad/s) of the driver?

A. 8.9 rad/s
B. 12.9 rad/s
C. 14.5 rad/s
D. 15.0 rad/s

13. An athlete throws a javelin with a velocity of 29.4 ms^{-1} at an angle of 41° relative to the horizontal ground. Neglecting air resistance, calculate how far the javelin lands from the launch position.

A. 133.1 m
B. 66.5 m
C. 70.4 m
D. 22.2 m

14. A particle of mass 0.2kg moves in a circular path with an angular velocity of 5 rad/s under the action of a centripetal force of 4N. What is the radius of the path?

A. 0.2 m
B. 0.4 m
C. 0.6 m
D. 0.8 m

15. A compact disc rotates from rest up to an angular speed of 30.0 rad/s in a time of 2.0s. What is the acceleration of the disc, assuming the angular acceleration is uniform?

A. 10 rad/s
B. 15 rad/s
C. 45 rad/s
D. 60 rad/s

16. An object in equilibrium has three forces, $F_1 = 30N$, $F_2 = 50N$ and $F_3 = 70N$, acting on it. The magnitude of the resultant of F_1 and F_2 will be

A 10N
B 20N
C 40N
D 70N

17. A car whose mass is 1200kg is accelerated from rest by a constant force of 2400N. What is the speed of the car after 8.0s?

A. 0.4ms^{-1}
B 1.6ms^{-1}
C 4.0ms^{-1}
D 16ms^{-1}

MYSTIC VENTURE

18. A 2kg mass is supported by a spring whose constant is 12N/cm. Calculate the extension produced in the spring.

A. 6.00 cm
B. 0.13 cm
C. 1.67 cm
D. 0.60 cm

19. A car of mass 1000kg hits a tree and decelerates at $50ms^{-2}$. What force does it exert on the tree?

A. 1000N
B. 5000N
C. 50000N
D 10000N

20. A 2 kg mass begins to slide down a plane when the inclination to the horizontal is α. The coefficient of friction of the surface of the plane is

A. $2g\sin\alpha$
B. $\tan\alpha$
C. $2\cos\alpha$
D. $2g\tan\alpha$

21. The unit of momentum is simply

A. N/m
B. Nm
C. $kgms^{-2}$
D. $kgms^{-1}$

22. Which of the following statement is true about collision?

A. in elastic collision, particles bounce off each other while losing energy
B. in perfectly inelastic collision, particles stick together while losing no energy
C. in inelastic collision, particles don't stick together while losing energy
D. none of the above

23. A car of mass 8000 kg moving at 10 ms^{-1} collides with a parked car of mass 2000 kg. If they stick together after collision, what is their common velocity after the collision?

A. $8ms^{-1}$
B. $18ms^{-1}$
C. $80ms^{-1}$
D. $0.8ms^{-1}$

24. Which of the following is not true concerning projectile motion?

A. The vertical motion is independent of acceleration due to gravity.
B. Horizontal and vertical motions are independent of each other.
C. Time taken to cover horizontal range is twice that of maximum height.
D. Trajectory of the motion is parabolic.

25. Two forces, 8N and 2N, act in opposite directions on a body of mass 10kg. Calculate the acceleration of the body.

A 1.6 ms^{-2}
B 1.0 ms^{-2}
C 0.80 ms^{-2}
D 0.6 ms^{-2}

26. A body of mass 20kg is to be given an acceleration of 10ms^{-2}. Calculate the force required when acceleration is vertically upwards.

A 1960 N
B 396 N
C 200 N
D 100 N

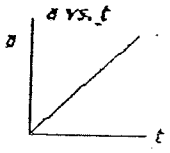
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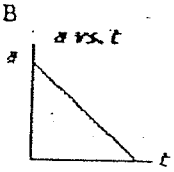
27. When a body is moving with a constant force, which of the following is correct?

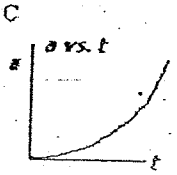
A. Its acceleration is constant
B. Its velocity is constant
C. Its momentum is constant
D. Its power is constant

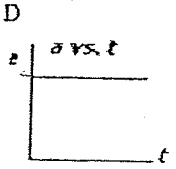
VENTURES

28. A ball is thrown vertically upward into the air. Which of the following acceleration-time graphs represents the ball's motion?

A 

B 

C 

D 

Use the information below to answer the two questions that follow it. .
A tennis ball is thrown vertically upwards with an initial velocity of 30 ms^{-1}.

29. What is the maximum height attained by the tennis ball?

A. 45.9 m
B. 30.0 m
C. 15.0 m
D. 294.0 m

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40. A box - shaped piece of iron has a cross-section of 15cm^2 and a height of 3cm. When a shearing force of 0.50N is applied to the upper surface, the upper surface displaces 4mm relative to the bottom surface. What is the shearing stress?

A. 300 Pa — 360Pa · A
B. 123 Pa — 333Pa · B.
C. 250 Pa — 250Pa · C
D. 430 Pa — 430Pa · D

41. An iron rod 4 m long and 0.5 cm^2 in cross section stretches 1 mm when a mass of 225kg is hung from its lower end. Compute Young's modulus for the iron.

A. 0.25×10^{-3} N/m^2 — 2.25×10^{11} N/m^2
B. 1.80×10^{4} N/m^2 — 1.80×10^{4} N/m^2
C. 4.41×10^{7} N/m^2 — 4.41×10^{7} N/m^2
D. 1.10×10^{-5} N/m^2 — 1.80×10^{11} N/m^2

42. A swimmer dives below the surface of a river. If the pressure acting on the surface equals the atmospheric pressure, calculate the pressure he experiences at a point 5.5 m below the surface.

A. 1.55×10^{5} Pa
B. 15.50×10^{5} Pa
C. 5.39×10^{3} Pa
D. 53.90×10^{3} Pa

43. The water pressure gauge indicates the pressure on the ground floor of a story building to be 300 kPa. If water is being pumped through pipes up the building, how high will it rise?

A. 0.31 m
B. 3.06 m
C. 30.6 m
D. 306 m

44. A periodic motion in which the acceleration of a body is proportional to its displacement and is directed towards a fixed point is known as

A. Circular motion
B. Rotational motion
C. Simple harmonic motion
D. Damped oscillatory motion

45. Which of the following statements is not true about viscosity?

A. It is the opposition to motion between solid surfaces.
B. The viscosities of liquids are generally significantly larger than those of gases.
C. Viscosities of liquids and gases depend on temperature.
D. The viscosity of a liquid decreases as temperature is increased.

46. From the kinetic theory of gases the total pressure P exerted on a cube-shaped box of side *l* and filled with molecules of an ideal gas can be expressed as:

A. $P = \frac{\rho \bar{C}^2}{3}$
B. $P = \frac{\rho \overline{C^2}}{3}$
C. $P = 3\rho \overline{C^2}$
D. $P = 3\rho \bar{C}^2$

47. An enclosure used to fill *He* balloons has a volume of 0.6 m^3 and contains 3.00 mol of *He* gas at 22 ^{0}C. If *He* behaves like an ideal gas, calculate the total translational KE of the molecules of the gas. *R* is 3.37 J/mol.K.

A. 1.111 kJ
B. 11.11 kJ
C. 111.1 kJ
D. 1111 kJ

48. An ideal gas at 17 ^{0}C has a pressure 1.01 x 10^5 Pa and is compresses isothermally until its volume us halved. Calculate the final temperature of the gas.

A. 8.5 ^{0}C
B. 1.7 ^{0}C
C. 17 ^{0}C
D. 34 ^{0}C

49. Which of the following statements is/are true?
I. An isotherm is a line on a P-V diagram that represents a constant-temperature process.
II. Boyle's law can be illustrated on an isotherm
III. Charles law cannot be represented on an isotherm

A. I only
B. I and II only
C. I and III only
D. I, II and III

50. In establishing a temperature scale, the temperature at which it is expected that a particular physical event always takes place is known as the

A. Boiling point
B. Fixed point
C. Triple point
D. Fundamental interval

51. The equivalent of 45 ^{0}C on the Fahrenheit scale will be

A. 7.22 ^{0}F
B. 72.2 ^{0}F
C. 11.3 ^{0}F
D. 113 ^{0}F

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52. Which of the following is NOT a thermometric property?

A. The expansion of a gas when heated
B. The contraction of a liquid when cooled
C. The melting of a solid when heated
D. The increase in the resistance of a conductor when heated

53. A thermometer in a research laboratory reads a room temperature of 68 °F. The equivalent of this temperature on the Kelvin scale is

A. 341 K
B. 205 K
C. 293 K
D. 280 K

54. A metal rod of length 100 cm is made of a material which has a thermal coefficient of linear expansion of 3.32×10^{-5}/°C. If the rod is heated from 35 °C to 45 °C, how much longer will it become?

A. 3.32×10^{-5} m
B. 3.32×10^{-4} m
C. 3.32×10^{-3} m
D. 3.32×10^{-2} m

55. A certain substance melts and freezes at +200°C. A block of this material of mass 500 g, is entirely solid at + 200°C. When heated, it takes 50,000 cal of energy to melt the substance entirely into liquid at + 200°C. What is the heat of fusion for this material?

A. 0.100 cal/g
B. 1.00 cal/g
C. 10.0 cal/g
D. 100 cal/g

56. The term *latent heat of vaporization* refers to the amount of heat

A. necessary to convert a certain amount of liquid matter to the gaseous state.
B. necessary to convert a certain amount of solid matter to the liquid state.
C. produced when a substance vaporizes.
D. needed to raise the temperature of a substance by 1 °C.

57. Which of the heat transfer mechanisms depends on the motion of mass from one region of space to another?

A. Conduction
B. Vaporisation
C. Convection
D. Radiation

58. Calorimetry is the science of

A. transferring heat
B. measuring heat
C. temperature measurements
D. weighing fluids

59. The principle of operation of a hydraulic press is based on

A. Charles law
B. Boyle's law
C. Pascal's law
D. Newton's law

60. A 50-kilogram fire-fighter is on a ladder 15 meters above the ground. When the fire-fighter descends to 10 meters above the ground, his gravitational potential energy will decrease by

A. 4900 joules
B. 7350 joules
C. 490 joules
D. 2450 joules

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